GHXTox: Peptide toxicity prediction by fusing a task-adaptive protein language model with pLDDT-aware multi-scale 3D structural modeling

LaLaLa, LaLaLa, LaLaLa, LaLaLa

Abstract

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer. However, toxicity assessment remains a critical bottleneck in peptide-based drug development. Existing methods have begun to explore multi-dimensional (multimodal) feature fusion and toxicity prediction based on the three-dimensional (3D) conformations of peptides, yet most of them still underutilize 3D structural information, and systematic multi-dimensional fusion remains rare. Here, we propose GHXTox, a hierarchical dual-level fusion framework. In the first level, a sequence-enhanced 3D structural expert integrates sequence semantics with spatial structural features through predicted local-distance difference test (pLDDT)-aware cross-attention. In the second level, this structural expert is fused with a task-adaptive one-dimensional (1D) sequence expert at the probability level, using weights frozen on training out-of-fold (OOF) predictions. The 1D branch analyzes sequences with a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) and a bidirectional Mamba state-space model. The 3D expert integrates an ESMFold-derived structural anchor, pLDDT-aware residue geometry, full-atom-derived chemical sites, and an availability-masked conformer-statistics residual for peptides with valid ensemble information; samples without conformer statistics fall back exactly to the single-structure pathway. On the frozen historical benchmark test sets, GHXTox achieves Test1 MCC of 0.8705 and AUPRC of 0.9674, and Test2 MCC of 0.6673 and AUPRC of 0.8046, while providing more detailed biological interpretations for peptide toxicity prediction.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware multi-scale 3D modeling; interpretability

1 INTRODUCTION

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer [1]. Peptide toxicity is a critical safety consideration throughout the drug development pipeline, and the accurate and efficient identification of toxic peptides remains a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable.

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional, 1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks, and a bidirectional long short-term memory network with residual connections, achieving strong

performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, such methods are constrained by the limited accuracy of current structure prediction models on short peptides and typically rely on a single predicted conformation; structural confidence, i.e., the predicted local distance difference test (pLDDT) score, is either discarded through hard filtering or ignored, and multi-conformer ensembles are not considered. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines a task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable hierarchical dual-level fusion framework for peptide toxicity prediction. In the first level, a sequence-enhanced 3D structural expert integrates sequence semantics with spatial structural features through pLDDT-aware cross-attention. In the second level, this structural expert is fused with a task-adaptive 1D sequence expert at the probability level, with weights frozen on training out-of-fold (OOF) predictions. In the 1D branch, a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) is combined with a bidirectional Mamba state-space model for sequence analysis [8–10]. In the 3D branch, ESMFold provides a common single-structure anchor for all peptides [11]. AlphaFold2/ColabFold, PepFlow, and BioEmu were also examined as conformer-generation references during structural development [12–15]; the retained default conformer residual uses only the prespecified PepFlow-covered subset. Gated sequence–structure cross-attention in the adjacent neuropeptide-prediction task further informs the interaction design, without transferring its labels or task-specific decision rule [16]. On this basis, we design a multi-scale structural information extraction strategy that progressively captures structural cues from local residue functional groups (chemical sites) to the whole peptide, and the pLDDT confidence output by structure prediction models is exploited through continuous down-weighting and dynamic gating of structural features. The 1D ensemble probability is first calibrated using training out-of-fold predictions, after which the calibrated 1D probability and the structural-expert probability are combined using OOF-frozen fusion weights. Ablation experiments demonstrate that the two dimensions complement each other effectively and that the fusion improves several fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2; interpretability analyses provide more detailed biological explanations for the predictions.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable hierarchical dual-level fusion framework that integrates a task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling; (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on the frozen historical benchmark test sets show that GHXTox improves several

fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2, and ablation, robustness, and interpretability analyses confirm the effectiveness and complementarity of the two dimensions.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as the benchmark Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional benchmark Test2 set, containing 46 toxic and 536 non-toxic peptides, is used for frozen historical evaluation. Both Test1 and Test2 had been observed during historical project development and are therefore treated as frozen historical benchmarks rather than newly blinded external validation sets; their labels were not used in the formal OOF calibration, fusion-weight selection, or threshold-freezing procedure. ToxinPred 3.0 [23] was consulted as an external method and data-source reference during project development, but its records were not added to the present training set or to the formal Test1/Test2 result tables. Table 1 summarizes the statistics of the datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Benchmark test
Test2	46	536	582	Benchmark test

2.2 Overview of the Model

Figure 1 shows an overview of GHXTox, a hierarchical dual-level fusion framework for peptide toxicity prediction. The framework consists of a task-adaptive one-dimensional (1D) sequence expert, a sequence-enhanced three-dimensional (3D) structural expert, and a probability fusion module driven by training out-of-fold (OOF) predictions. Within the structural expert, GHXTox first performs feature-level interaction between sequence semantics and spatial structural information through pLDDT-aware cross-attention. At the top level of the model, the prediction probabilities produced by the 1D sequence expert and the 3D structural expert are then fused at the probability level. GHXTox therefore combines intra-expert feature-level fusion with inter-expert probability-level fusion, preserving the independent discriminative capability of each branch while exploiting complementary information from amino acid sequences, spatial geometry, local chemical environments, and conformational variability.

Figure 1: Overall architecture of GHXTox. GHXTox adopts a hierarchical dual-level fusion design. The task-adaptive 1D sequence expert extracts toxicity-related sequence features with a LoRA-fine-tuned ProtT5 encoder and a bidirectional Mamba network, followed by three-seed ensembling and out-of-fold (OOF) Platt calibration to produce p_{1D}. The sequence-enhanced 3D structural expert extracts multi-scale structural information through an ESM2-enhanced sequence encoder, a pLDDT-aware equivariant graph neural network, full-atom chemical-site modeling, and an availability-masked conformer residual branch. Within this expert, sequence and spatial residue representations are integrated through pLDDT-aware cross-attention to produce p_{SE3D}. The two expert probabilities are fused with frozen weights of 0.72 and 0.28, and the final toxicity decision is obtained with a threshold of 0.4702225653.

2.2.1 Task-Adaptive 1D Sequence Expert

The task-adaptive 1D sequence expert takes the amino acid sequence of a peptide as input. After tokenization, the sequence is processed by the ProtT5 protein language model [8] to obtain contextualized, residue-level representations. To adapt this general-purpose protein language model to peptide toxicity prediction, we employ low-rank adaptation (LoRA) [9] for parameter-efficient fine-tuning, learning task-specific updates while the majority of the pretrained backbone parameters remain frozen. The resulting residue-level representations are then fed into a bidirectional Mamba state-space network [10], which models residue dependencies in both the N-to-C and C-to-N directions. After residual connections, layer normalization, and masked pooling, a classification head produces the sequence-level toxicity probability. To reduce the variance arising from random initialization, we train three models on the full training set with different random seeds and average their output probabilities. The ensemble probability is then calibrated with Platt scaling [24] fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability p_{1D}.

2.2.2 Sequence-Enhanced 3D Structural Expert

The sequence-enhanced 3D structural expert comprises four subbranches: an internal sequence-semantic subbranch, a spatial-geometric subbranch, a full-atom chemical-site subbranch, and an availability-masked conformer-ensemble residual subbranch. Collectively, these components cover five complementary representation scales: residue-level semantics, atom-level chemical sites, spatial residue neighborhoods, peptide-level global properties, and conformer-ensemble statistics. In the sequence-semantic subbranch, amino acid residue types, functional-group properties, explicit physicochemical descriptors, and ESM2 residue embeddings [11] are projected into a shared feature space and then processed by a two-layer Transformer encoder [25] to generate contextualized sequence representations. In parallel, the spatial-geometric subbranch constructs a hybrid residue graph that combines sequential adjacency edges with spatial-neighborhood edges derived from the predicted peptide structure. Residue physicochemical properties, local geometric descriptors, ESM2 embeddings, three-dimensional coordinates, and residue-level pLDDT scores are then fed into three pLDDT-aware equivariant graph neural network (EGNN) [26] layers to update the residue states and spatial representations.

To characterize local chemical environments that cannot be fully captured by a Cα-based residue graph alone, we extract chemical sites corresponding to charged, aromatic, and hydrophobic groups as well as hydrogen-bond donors and acceptors from the full-atom peptide structure. Message passing [27] is then performed over these sites using pairwise distances, directional relationships, interaction types, and structural confidence. The resulting chemical-site features are mapped back into the corresponding residue-level representations through a zero-initialized residual projection. This design allows the model to progressively learn toxicity-related local spatial–chemical patterns while preserving the initial geometric backbone. For peptides with

available conformer ensembles, the model further extracts residue-level root-mean-square fluctuation (RMSF), pairwise distance variance, contact occupancy under multiple distance cutoffs, and peptide-level root-mean-square deviation (RMSD) and radius of gyration. These statistics are incorporated through an availability-masked residual branch to refine the structural expert's output. When conformer-ensemble information is unavailable, this residual contribution is constrained to exactly zero.

Within the structural expert, the contextualized sequence and confidence-aware spatial representations are integrated through pLDDT-aware cross-attention, followed by peptide-level confidence gating; the detailed construction is given in Section 2.4.6. After masked mean pooling, fusion with global peptide-level physicochemical features, and multilayer perceptron (MLP) classification, the sequence-enhanced structural expert obtains a base structural logit. The base structural logit is further corrected by an availability-masked, zero-initialized conformer residual before being transformed into the structural-expert probability p_{SE3D}.

2.2.3 Hierarchical Fusion and Toxicity Decision

At the top level of the model, the calibrated 1D sequence expert and the sequence-enhanced 3D structural expert are not combined through further intermediate-feature concatenation; instead, their outputs are fused with fixed weights in the probability space. The final toxicity probability is defined as

$$p_{\mathrm{GHXTox}} = 0.72\, p_{\mathrm{1D}} + 0.28\, p_{\mathrm{SE3D}}. \tag{1}$$

The larger sequence-expert weight is an empirical outcome of the OOF fusion search rather than a manually assigned prior; the structural expert provides complementary evidence on spatial geometry, local chemical environments, and conformational dynamics. The fusion weights are determined exclusively from group-aware training out-of-fold (OOF) predictions and are frozen before evaluation on the frozen historical benchmark test sets. In the implementation, the two expert probabilities are likewise combined linearly, without re-optimizing the fusion parameters on the test data.

Finally, the fused probability is converted into a binary toxicity prediction using a fixed decision threshold:

$$\hat{y} = \begin{cases} 1, & p_{\mathrm{GHXTox}} \geq 0.4702225653, \\ 0, & p_{\mathrm{GHXTox}} < 0.4702225653, \end{cases} \tag{2}$$

where $\hat{y} = 1$ denotes a toxic peptide and $\hat{y} = 0$ denotes a non-toxic peptide. The probability-calibration parameters, expert-fusion weights, and classification threshold are all determined solely from training out-of-fold (OOF) predictions. The current procedure does not use Test1 or Test2 labels to fit these quantities. Because both sets had nevertheless been observed during historical project development, their results are reported as frozen retrospective benchmarks rather than newly blinded external validation. This hierarchical design enables fine-grained sequence–structure interaction through pLDDT-aware cross-attention inside the structural expert, while the frozen probability-level fusion at the top level enhances inter-expert complementarity and predictive stability.

2.3 Task-Adaptive 1D Sequence Modeling

The 1D sequence expert is designed to learn toxicity-related contextual semantics and sequential dependencies directly from peptide amino acid sequences. Instead of reducing a frozen protein-language-model output to a fixed mean vector, GHXTox preserves the residue-level representations generated by ProtT5 and adapts the pretrained encoder to the peptide toxicity prediction task through low-rank adaptation (LoRA). The resulting residue-level representations are then processed by a bidirectional Mamba state-space encoder to jointly model local residue patterns, N-to-C dependencies, and C-to-N dependencies. The encoded residue features

are aggregated via masked mean pooling and passed to a linear classification head to produce the single-model toxicity probability. Finally, three models trained on the full training set with different random seeds are combined by equal-weight probability averaging, followed by Platt calibration fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability p_{1D}.

This design organizes pretrained protein semantics, parameter-efficient task adaptation, bidirectional sequence modeling, and probability calibration into a coherent processing pipeline. While most of the pretrained knowledge in ProtT5 is retained, LoRA enables task-specific adaptation of the residue representations. The bidirectional Mamba encoder further reorganizes these representations according to toxicity-related sequential patterns, whereas multi-seed ensembling and out-of-fold calibration reduce training variance and align the calibrated probability scale with that of the downstream structural expert.

2.3.1 Residue Tokenization and LoRA-Adapted ProtT5

Given a peptide sequence of length L, we first convert its amino acid residues into the space-separated format required by the ProtT5 tokenizer and map them to token indices. Special tokens are inserted automatically, and sequences of different lengths are padded to a uniform length within each batch. The maximum token length is set to 128, which is sufficient for all peptides used in this study. Under the current ProtT5 tokenizer configuration, the first L token positions correspond to the L amino acid residues, followed by the EOS token and, when required, padding tokens. Accordingly, the implementation constructs the residue-validity mask directly from the original peptide length:

$$\mathcal{S} = (a_1, a_2, \ldots, a_L), \qquad m_i = \begin{cases} 1, & 1 \le i \le L, \\ 0, & i > L. \end{cases} \tag{3}$$

where $m_i = 1$ denotes an actual amino acid position and $m_i = 0$ denotes a special or padding position; the same mask is applied in the Mamba scans and the subsequent global pooling.

The tokenized sequence is then processed by the ProtT5-XL-UniRef50 encoder [8] to generate contextualized residue-level representations:

$$\mathbf{H}^{\mathrm{PLM}} = \mathrm{ProtT5}_{\mathrm{LoRA}}(\mathcal{S}), \qquad \mathbf{H}^{\mathrm{PLM}} \in \mathbb{R}^{L \times 1024}. \tag{4}$$

Retaining the complete residue-level representation matrix enables the downstream encoder to preserve position-specific contextual semantics. Each residue vector thus encodes not only its amino acid identity but also its conditional context within the complete peptide sequence.

To adapt the general-purpose ProtT5 encoder to the peptide toxicity prediction task, we apply low-rank adaptation (LoRA) [9] to the linear transformations within the encoder. For an original linear layer $\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b}$, the pretrained weight matrix $\mathbf{W}_0$ is frozen, and the task-specific update is factorized into two trainable low-rank matrices:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \qquad \mathbf{A} \in \mathbb{R}^{r \times d_{\mathrm{in}}}, \quad \mathbf{B} \in \mathbb{R}^{d_{\mathrm{out}} \times r}. \tag{5}$$

The LoRA-augmented forward pass is therefore

$$\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b} + \frac{\alpha}{r}\mathbf{B}\mathbf{A}\,\mathrm{Dropout}(\mathbf{x}). \tag{6}$$

We set the LoRA rank to $r = 8$, the scaling coefficient to $\alpha = 16$, and the LoRA dropout rate to 0.1, yielding a scaling factor of $\alpha/r = 2$. Matrix $\mathbf{A}$ is initialized with Kaiming initialization, whereas $\mathbf{B}$ is initialized to zero; consequently, the LoRA branch initially leaves the pretrained transformation unchanged and progressively learns toxicity-specific updates during training.

Unlike conventional configurations that apply LoRA only to selected attention projections, the current implementation augments all linear layers within the ProtT5 encoder. The pretrained

base transformations and embeddings remain frozen, whereas the low-rank parameters and the downstream sequence- classification modules are trainable.

The 1024-dimensional ProtT5 representations are then projected into the 192-dimensional hidden space used by the Mamba encoder [10]:

$$\mathbf{X}^{(0)} = \mathrm{Dropout}\left(\mathrm{GELU}\left[\mathrm{LN}\left(\mathbf{H}^{\mathrm{PLM}}\mathbf{W}_p + \mathbf{b}_p\right)\right]\right), \qquad \mathbf{X}^{(0)} \in \mathbb{R}^{L\times 192}. \tag{7}$$

where LN denotes layer normalization. The projection block consists of a linear transformation, layer normalization, GELU activation, and dropout. It reduces the computational dimension for the downstream state-space encoder while providing a task-specific nonlinear transformation of the ProtT5 residue representations.

2.3.2 Bidirectional Mamba Sequence Encoder

Although ProtT5 provides contextualized residue representations, these representations primarily encode general protein semantics learned during large-scale pretraining. To further model the sequential dependencies specifically associated with peptide toxicity, two bidirectional Mamba layers are placed after the ProtT5 projection block. Mamba employs an input-dependent selective state-space mechanism that dynamically controls which residue information is retained, updated, or suppressed during sequential scanning [10].

For a unidirectional Mamba scan, the input representation $\mathbf{X} \in \mathbb{R}^{L\times d_{\mathrm{model}}}$ with $d_{\mathrm{model}} = 192$ is first projected into a content branch and a gating branch:

$$[\mathbf{U}, \mathbf{Z}] = \mathbf{X}\mathbf{W}_{\mathrm{in}} + \mathbf{b}_{\mathrm{in}}, \qquad \mathbf{U}, \mathbf{Z} \in \mathbb{R}^{L\times d_{\mathrm{inner}}}, \quad d_{\mathrm{inner}} = 2d_{\mathrm{model}} = 384, \tag{8}$$

i.e., the projection expands the feature dimension from 192 to 768 and splits it into two branches, each of dimension $d_{\mathrm{inner}} = 384$. The content branch is processed by a depthwise one-dimensional convolution with a kernel width of 4, followed by a SiLU activation:

$$\widetilde{\mathbf{U}} = \mathrm{SiLU}(\mathrm{DWConv1D}(\mathbf{U})), \qquad \widetilde{\mathbf{U}}^{\mathrm{m}}_t = m_t\widetilde{\mathbf{U}}_t, \tag{9}$$

where $\widetilde{\mathbf{U}}^{\mathrm{m}}_t$ denotes the convolved feature after applying the residue-validity mask. This local convolution extracts short-range residue patterns before the state-space scan, enabling the module to combine local sequence motifs with the longer-range context accumulated through state recurrence. Input-dependent projections then generate the discretization step size and the state-space parameters from the masked feature:

$$[\delta_t, \mathbf{B}_t, \mathbf{C}_t] = \mathrm{Linear}(\widetilde{\mathbf{U}}^{\mathrm{m}}_t), \qquad \Delta_t = \mathrm{Softplus}(\mathbf{W}_\Delta\delta_t + \mathbf{b}_\Delta). \tag{10}$$

The state-transition matrix is parameterized to be negative and is discretized according to the input-dependent step size:

$$\mathbf{A} = -\exp(\mathbf{A}_{\log}), \qquad \bar{\mathbf{A}}_t = \exp(\Delta_t\mathbf{A}), \qquad \bar{\mathbf{B}}_t = \Delta_t\mathbf{B}_t, \tag{11}$$

where $\mathbf{A}_{\log} \in \mathbb{R}^{d_{\mathrm{inner}}\times d_{\mathrm{state}}}$ is a learnable parameter matrix. The selective state recurrence is expressed as

$$\mathbf{S}_t = \bar{\mathbf{A}}_t \odot \mathbf{S}_{t-1} + \bar{\mathbf{B}}_t \odot \widetilde{\mathbf{U}}^{\mathrm{m}}_t, \qquad Y_t = \sum_{n=1}^{d_{\mathrm{state}}} S_{t,n}C_{t,n}, \tag{12}$$

where the state dimension is set to $d_{\mathrm{state}} = 16$. The state-space output at position t is subsequently combined with the gating branch and the skip term:

$$\mathbf{O}_t = \mathbf{W}_{\mathrm{out}}\left[\mathbf{Y}_t \odot \mathrm{SiLU}(\mathbf{Z}_t) + \mathbf{D} \odot \widetilde{\mathbf{U}}^{\mathrm{m}}_t\right]. \tag{13}$$

At invalid positions, $\widetilde{\mathbf{U}}^{\mathrm{m}}_t$ is zero and Δ_t is also zeroed, preventing padding tokens from injecting information into the recurrent states.

A unidirectional scan accesses only one side of the sequence context, whereas peptide toxicity may depend on residue patterns associated with both termini and with interactions spanning different sequence positions. Therefore, in the ℓ-th bidirectional Mamba layer, forward and backward scans are applied to the normalized sequence using the residue-validity mask $\mathbf{m}$:

$$\begin{aligned}\mathbf{F}^{(\ell)} &= \mathcal{M}_f^{(\ell)}(\mathrm{LN}(\mathbf{X}^{(\ell)}), \mathbf{m}),\\ \mathbf{R}^{(\ell)} &= \mathrm{Reverse}\left[\mathcal{M}_b^{(\ell)}\left(\mathrm{Reverse}\left[\mathrm{LN}(\mathbf{X}^{(\ell)})\right], \mathrm{Reverse}(\mathbf{m})\right)\right],\end{aligned} \tag{14}$$

where the backward-scan output is restored to the original residue order. The two directional outputs, each of dimension 192, are concatenated along the feature dimension (384 in total) and fused by a linear layer that projects the representation back to 192 dimensions:

$$\mathbf{M}^{(\ell)} = \mathrm{GELU}\left(\mathrm{Dropout}\left(\mathbf{W}_m\left[\mathbf{F}^{(\ell)} \parallel \mathbf{R}^{(\ell)}\right] + \mathbf{b}_m\right)\right), \qquad \mathbf{X}^{(\ell+1)} = \mathbf{X}^{(\ell)} + \mathrm{Dropout}(\mathbf{M}^{(\ell)}), \tag{15}$$

where $\parallel$ denotes feature concatenation, the dropout inside the GELU block is applied within the fusion module, and the second dropout is applied before the residual addition. Two bidirectional Mamba layers are stacked in the final model. The first layer reconstructs bidirectional residue dependencies from the task-adapted ProtT5 features, whereas the second layer further aggregates the resulting contextual representations. This design allows each residue position to receive conditional information from both its N-terminal and C-terminal contexts.

2.3.3 Masked Pooling and Class-Weighted Classification

After two bidirectional Mamba layers, the model produces residue-level sequence representations:

$$\mathbf{X}^{(2)} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L], \qquad \mathbf{X}^{(2)} \in \mathbb{R}^{L\times 192}. \tag{16}$$

Because peptide sequences have variable lengths, GHXTox does not rely on a fixed-position classification token. Instead, masked mean pooling is applied over the valid amino acid positions:

$$\mathbf{h}_{1D} = \frac{\sum_{i=1}^{L} m_i \mathbf{x}_i}{\sum_{i=1}^{L} m_i}, \tag{17}$$

where the denominator equals the number of valid residue positions. This operation converts the variable-length residue sequence into a fixed 192-dimensional peptide-level representation while ensuring that EOS and padding positions do not contribute to the pooled feature. In contrast to global max pooling, masked mean pooling aggregates distributed evidence across the entire peptide and avoids a representation dominated by a single extreme activation.

The peptide-level representation is first normalized and then passed through a single-output linear classification head:

$$z_{1D} = \mathbf{w}_c^{\top}\,\mathrm{LN}(\mathbf{h}_{1D}) + b_c, \qquad p_{1D}^{\mathrm{raw}} = \sigma(z_{1D}) = \frac{1}{1+\exp(-z_{1D})}, \tag{18}$$

where z_{1D} denotes the 1D expert logit and p_{1D}^{raw} the corresponding uncalibrated toxicity probability. The classification head of the current 1D expert consists of layer normalization followed by a single linear layer, rather than a multilayer perceptron; it is therefore referred to as a linear classification head throughout the architecture and Methods sections.

Because non-toxic peptides substantially outnumber toxic peptides in the training data, a positive-class-weighted binary cross-entropy loss is used in the current 1D expert instead of SMOTE or Borderline-SMOTE. For a training partition containing N_+ toxic and N_- non-toxic peptides, the positive-class weight is defined as

$$w_+ = \frac{N_-}{N_+}, \tag{19}$$

and the class-weighted objective is expressed as

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N}\sum_{j=1}^{N}\left[w_{+}y_j \log \sigma(z_j) + (1-y_j)\log(1-\sigma(z_j))\right]. \tag{20}$$

During five-fold out-of-fold training, w_+ is calculated independently from the class counts of each fold's training partition; for full-data training, it is recalculated using the complete training set. This strategy increases the contribution of the minority toxic class without generating synthetic peptide sequences or feature vectors, while preventing any information from the held-out validation fold from entering the training procedure.

2.3.4 Three-Seed Ensemble and OOF Platt Calibration

The predictions of deep neural networks may vary because of parameter initialization, mini-batch ordering, and stochastic optimization. To reduce the variance associated with an individual training run, three 1D expert models are trained under identical data, architecture, and optimization settings using the random seeds 42, 123, and 2025. The deployed ensemble consists of three models trained on the complete training set, rather than the fold-specific checkpoints produced during cross-validation.

For sample j, the toxicity probability produced by the model trained with seed k is denoted by $p_j^{(k)} = \sigma(z_j^{(k)})$, $k \in \{1,2,3\}$, and the uncalibrated ensemble score is obtained by equal-weight arithmetic averaging:

$$s_j = \frac{1}{3}\sum_{k=1}^{3} p_j^{(k)}. \tag{21}$$

No individual seed is selected according to its performance on Test1 or Test2, and no seed-specific ensemble weight is introduced. All members share the same architecture and training protocol; the different random seeds provide averaging over a limited set of initialization and optimization stochasticities.

Although probability averaging reduces prediction variance, neural-network probabilities may remain systematically overconfident or underconfident. Moreover, the top-level GHXTox framework subsequently combines the 1D expert with the structural expert, making probability-scale compatibility particularly important. Therefore, Platt scaling [24] is applied after the three-model probability average.

Training out-of-fold predictions are first generated using the fixed group-aware five-fold partition. Each training peptide receives an OOF probability from a model trained without the fold containing that peptide. The OOF probabilities from the three random seeds are then averaged to obtain the OOF ensemble score s_j^{OOF}. A two-parameter logistic calibrator is fitted using only these training OOF scores and their labels:

$$p_{1D,j} = \sigma\left(a\,\mathrm{logit}\left(\widetilde{s}_j\right) + b\right), \tag{22}$$

where $\mathrm{logit}(s) = \ln(s/(1-s))$, $\widetilde{s}_j = \mathrm{clip}(s_j, \varepsilon, 1-\varepsilon)$ with $\varepsilon = 10^{-6}$, and a and b denote the Platt calibration parameters; the clip prevents infinite logits for probabilities of exactly 0 or 1, and the same clipping is applied to the OOF scores used in fitting. The parameters are estimated as

$$(a,b) = \arg\min_{a,b} \sum_{j=1}^{N} \mathrm{BCE}\left[y_j,\ \sigma\left(a\,\mathrm{logit}\left(\widetilde{s}_j^{\mathrm{OOF}}\right) + b\right)\right]. \tag{23}$$

The scaling parameter is reparameterized as $a = \mathrm{clip}(\exp(\eta),\, 0.02, 50) > 0$, which guarantees a monotonic probability transformation and therefore preserves the sample ranking of the ensemble score; AUROC and AUPRC are consequently unchanged by calibration. For benchmark test

or previously unseen peptides, the output probabilities of the three full-data models are first averaged, after which the frozen calibration parameters are applied:

$$p_{1\mathrm{D}} = \sigma\left(a\,\mathrm{logit}\left(\mathrm{clip}\left(\frac{1}{3}\sum_{k=1}^{3} p^{(k)},\ \varepsilon,\ 1-\varepsilon\right)\right) + b\right). \tag{24}$$

The labels of Test1 and Test2 are not used for ensemble-member selection, calibration fitting, or 1D threshold optimization. The calibrated probability $p_{1\mathrm{D}}$ is the sequence-expert output used in the top-level probability fusion described in Section 2.2.3. The complete order of operations is therefore

3-seed probabilities $\rightarrow$ averaging $\rightarrow$ OOF-fitted Platt calibration $\rightarrow p_{1\mathrm{D}} \rightarrow$ 1D/3D fusion.

Calibration is performed within the 1D expert before the top-level 1D/3D fusion; no additional Platt calibrator is fitted after the final GHXTox probability fusion.

2.4 pLDDT-Aware Multi-Scale 3D Structural Modeling

Peptide toxicity is influenced not only by amino acid composition and linear residue order but also by the three-dimensional (3D) arrangement of charged groups, hydrophobic regions, aromatic systems, and sulfur-containing moieties. However, peptides are generally short, conformationally flexible, and poorly covered by experimentally determined structures. Treating a single predicted conformation as an error-free structural observation may therefore amplify structure-prediction errors. Conversely, representing a peptide with only Cα coordinates or peptide-level summary descriptors fails to capture the local spatial-chemical environments that are relevant to toxicity.

To address these limitations, GHXTox employs a pLDDT-aware multi-scale structural expert that captures complementary information at five levels: residue semantics, spatial residue neighborhoods, full-atom chemical sites, peptide-level global properties, and conformer-ensemble statistics. The model first extracts residue-level Cα coordinates, full-atom coordinates, and normalized pLDDT confidence scores from the predicted structure. An internal sequence-semantic subbranch combines amino acid residue types, functional-group categories, explicit physicochemical descriptors, and ESM2 residue embeddings to generate contextualized sequence states. In parallel, a spatial-geometric subbranch constructs a hybrid residue graph containing both sequential adjacency edges and spatial-neighborhood edges and propagates geometric information through pLDDT-aware equivariant graph layers.

To overcome the limited chemical resolution of a Cα-only residue graph, GHXTox additionally extracts typed chemical sites from the full-atom structure and performs message passing using dual-scale distances, directional relationships, interaction-compatibility channels, and structural confidence. For peptides with available conformer ensembles, an additional statistical residual branch summarizes residue fluctuations, distance variation, and contact occupancy. The residual is constrained to exactly zero when conformer information is unavailable.

Rather than discarding low-confidence structures through hard filtering, GHXTox treats pLDDT as a continuous reliability signal and incorporates it into geometric edge weighting, chemical-site messages, residue-level structural gating, and peptide-level structural gating. Low-confidence structural evidence is therefore down-weighted rather than removed, preserving structural coverage while limiting the influence of uncertain coordinates. Importantly, pLDDT is interpreted here as a local confidence estimate produced by the structure-prediction model, rather than an experimentally determined measure of structural accuracy, conformational flexibility, or biological activity. The detailed architecture of the structural expert is illustrated in Figure 2.

Figure 2: Detailed architecture of the sequence-enhanced structural expert. The structural expert integrates ESM2-enhanced residue semantics, a pLDDT-aware geometric residue graph, full-atom chemical-site message passing, and availability-masked conformer-ensemble statistics. Residue-level structural states are continuously modulated by node-level confidence before serving as keys and values in sequence–structure cross-attention. A peptide-level confidence gate further controls the fused structural context, while conformer statistics provide a zero-initialized logit residual only when valid ensemble information is available.

2.4.1 Peptide Structure Prediction and Conformer Preparation

For a peptide sequence of length L, $S = (a_1, a_2, \ldots, a_L)$, ESMFold is first used to predict its three-dimensional structure [11]. The predicted structure is stored as a full-atom PDB file, from which residue-level Cα coordinates, backbone coordinates, full-atom-derived chemical-site coordinates, and pLDDT scores are extracted. This operation is denoted as

$$(\mathbf{X}, \mathcal{A}, \mathbf{q}) = \mathrm{ESMFold}(\mathcal{S}), \qquad \mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L\times 3}, \quad \mathbf{q} = (q_1, \ldots, q_L) \in [0,1]^L, \tag{25}$$

where $\mathbf{X}$ contains the Cα coordinates, $\mathcal{A}$ denotes the full-atom-PDB-derived backbone and chemical-site geometry, and $\mathbf{q}$ contains the residue-level pLDDT confidence scores rescaled to $[0, 1]$. The Cα coordinates serve as the primary spatial variables of the residue-level equivariant graph, whereas the full-atom coordinates are used to construct local chemical sites. Before geometric encoding, the coordinates are centered at the peptide centroid to remove the effect of absolute translation.

A single predicted structure provides a common structural anchor but cannot represent potential conformational heterogeneity. Therefore, the current default GHXTox configuration additionally uses conformer-ensemble statistics on a prespecified subset of peptides covered by PepFlow [14]. Up to eight validated and aligned conformers are retained for each available peptide, with residue indices matched to a common sequence order. The PepFlow pilot250 subset covers 111 toxic and 139 non-toxic peptides with sequence lengths of 11–15 residues and up to eight conformers per peptide. The ensemble is represented as

$$\mathcal{C} = \{\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(M)}\}, \qquad 1 \le M \le 8, \quad \mathbf{X}^{(m)} \in \mathbb{R}^{L\times 3}, \tag{26}$$

where $\mathbf{X}^{(m)}$ denotes the Cα coordinates of conformer m. Rigid-body alignment is performed before the ensemble statistics are calculated, ensuring that the resulting variation reflects internal geometric changes rather than arbitrary global rotations or translations.

Because conformer ensembles are not available for every training peptide, a binary availability indicator is defined as

$$c_{\mathrm{conf}} = \begin{cases} 1, & \text{if valid ensemble statistics are available,} \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

When $c_{\mathrm{conf}} = 0$, the conformer-statistics tensors are filled with zeros, and the corresponding residual contribution is explicitly suppressed by the availability indicator. Missing ensemble information is therefore not interpreted as evidence of structural rigidity; instead, the model falls back exactly to the single-structure pathway. No conformer ensembles are available for Test1 or Test2 in the current evaluation assets; hence, their conformer-residual contribution is exactly zero.

The generated conformers are computational samples rather than experimentally determined trajectories or rigorous thermodynamic ensembles. Consequently, RMSF, contact occupancy, and distance variance are interpreted as sampling-derived descriptors of conformational variation rather than direct measurements of peptide dynamics in solution.

2.4.2 Residue Feature Construction and Hybrid Geometric Graph

The structural expert constructs both an internal sequence representation and a spatial-geometric representation at the residue level. Importantly, the internal sequence-semantic subbranch described here uses frozen ESM2 residue embeddings and a lightweight Transformer. It is distinct from the LoRA-ProtT5 and bidirectional Mamba expert described in Section 2.3. The former supports residue-level sequence–structure alignment inside the structural expert, whereas the latter acts as an independent top-level 1D prediction expert.

For residue i, the input to the internal sequence-semantic subbranch is

$$\mathbf{u}_i^{\mathrm{seq}} = \left[\mathbf{e}_i^{\mathrm{AA}} \parallel \mathbf{e}_i^{\mathrm{group}} \parallel \mathbf{f}_i^{\mathrm{res}} \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{28}$$

where $\mathbf{e}_i^{\mathrm{AA}} \in \mathbb{R}^{32}$ is the amino acid category embedding, $\mathbf{e}_i^{\mathrm{group}} \in \mathbb{R}^{12}$ is the functional-group category embedding, $\mathbf{f}_i^{\mathrm{res}} \in \mathbb{R}^{42}$ denotes the explicit residue-level physicochemical descriptors, and $\mathbf{e}_i^{\mathrm{ESM2}} \in \mathbb{R}^{1280}$ is the ESM2-650M residue representation; the projection P_{ESM} maps the ESM2 representation to 128 dimensions. Stacking the residue-level feature vectors yields the sequence-feature matrix $U_{\mathrm{seq}} \in \mathbb{R}^{L\times 214}$, where $214 = 32+12+42+128$. After a linear projection and positional encoding, the sequence is processed by a two-layer Transformer encoder with hidden dimension 128, four attention heads, and a feed-forward dimension of 512:

$$\mathbf{H}^{\mathrm{seq}} = \mathrm{Transformer}_2\left[P_{\mathrm{seq}}\left(U_{\mathrm{seq}}\right) + E_{\mathrm{pos}}\right], \qquad \mathbf{H}^{\mathrm{seq}} \in \mathbb{R}^{L\times 128}, \tag{29}$$

This representation encodes the contextual sequence semantics of each residue and subsequently serves as the query representation in the sequence–structure cross-attention module.

In the spatial subbranch, the initial representation of residue i is constructed as

$$\mathbf{h}_i^{(0)} = P_{\mathrm{sp}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel \mathbf{g}_i^{\mathrm{geo}} \parallel q_i \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{30}$$

where $\mathbf{g}_i^{\mathrm{geo}} \in \mathbb{R}^{16}$ is a lightweight geometric descriptor derived from the Cα coordinates and pLDDT scores. The 16-dimensional descriptor includes (i) residue-level pLDDT and local confidence statistics; (ii) radial position, contact fractions, and soft spatial density; (iii) nearest-neighbor, nonlocal-neighbor, and local sequential distances; and (iv) sine/cosine encodings of the Cα-trace angles and torsions. These explicit descriptors provide stable local geometric priors without introducing an overly complex handcrafted representation for short and uncertain predicted peptide structures.

GHXTox then constructs a hybrid residue graph:

$$\mathcal{G} = \left(\mathcal{V}, \mathcal{E}_{\mathrm{seq}} \cup \mathcal{E}_{\mathrm{knn}}\right), \tag{31}$$

where each residue corresponds to one node. Sequential edges $\mathcal{E}_{\mathrm{seq}}$ connect adjacent residues and preserve peptide-chain continuity, whereas $\mathcal{E}_{\mathrm{knn}}$ connects each residue to its 10 nearest spatial neighbors and is symmetrized. The hybrid graph therefore captures both covalent sequence order and nonlocal spatial contacts while avoiding large variations in graph density that may arise from a single fixed-distance cutoff. Equivalently, $A_{ij} \in \{0,1\}$ denotes whether residues i and j are connected by a sequential or k-nearest-neighbor edge in the hybrid graph.

For residues i and j, the Cα distance is

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2, \qquad \phi_k(d_{ij}) = \exp\left[-\gamma\left(d_{ij} - \mu_k\right)^2\right], \quad \gamma = \frac{1}{(\mu_{k+1} - \mu_k)^2}, \quad k = 1, \ldots, 16, \tag{32}$$

where the centers μ_k are uniformly spaced over the range 0–24 Å. The base edge representation additionally includes the pairwise structural confidence and the normalized sequence separation:

$$c_{ij} = \sqrt{q_i q_j}, \qquad s_{ij} = \frac{\log(1 + |i-j|)}{\log(L+1)}. \tag{33}$$

Thus, sequentially adjacent residues receive small values, whereas residues separated by longer sequence intervals approach one. The resulting edge feature is therefore $\mathbf{e}_{ij} = [\boldsymbol{\phi}(d_{ij}) \parallel c_{ij} \parallel s_{ij}]$. This formulation allows the model to distinguish sequentially adjacent contacts, long-range spatial contacts, and contacts supported by different levels of structural confidence.

2.4.3 pLDDT-Aware Equivariant Graph Encoding

To learn three-dimensional representations without relying on a particular global coordinate frame, the spatial branch employs three E(3)-equivariant graph neural network layers [26]. The network updates residue features using rotation- and translation-invariant quantities while updating coordinates equivariantly. Consequently, a global rigid transformation of the peptide structure does not change the predicted toxicity, whereas intermediate coordinate representations transform consistently with the input.

At layer ℓ, the message from residue j to residue i is computed as

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e^{(\ell)} \left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell)} \right], \tag{34}$$

where $\phi_e^{(\ell)}$ is a message multilayer perceptron. To prevent uncertain or distant contacts from contributing as strongly as local high-confidence contacts, a continuous edge weight is defined as

$$w_{ij}^{(\ell)} = A_{ij} \, \exp\left(-\frac{d_{ij}^{(\ell)}}{12} \right) \left[q_{\min} + (1 - q_{\min}) \, c_{ij}^{\rho} \right], \qquad q_{\min} = 0.1, \quad \rho = 1.5, \tag{35}$$

The term $\exp(-d_{ij}^{(\ell)}/12)$ imposes a smooth distance decay with a characteristic decay length of 12 Å, and both the distance and the edge weight are recomputed at each layer from the current coordinates. Here, $A_{ij} \in \{0, 1\}$ is the adjacency indicator of the hybrid graph defined in Section 2.4.2, and c_{ij} is the pairwise structural confidence defined in Section 2.4.2. The minimum weight $q_{\min}$ prevents low-confidence regions from being completely disconnected, whereas the exponent ρ increases the relative contribution of high-confidence contacts. The distance-decay term further reduces the influence of more distant spatial neighbors.

Messages received by node i are aggregated with a normalization floor:

$$\mathbf{m}_i^{(\ell)} = \frac{\sum_j w_{ij}^{(\ell)} \, \mathbf{m}_{ij}^{(\ell)}}{\max\left(\sum_j w_{ij}^{(\ell)}, \, 1 \right)}. \tag{36}$$

The aggregated message is used to update the node representation with a residual connection followed by layer normalization:

$$\mathbf{h}_i^{(\ell+1)} = \mathrm{LN}\left\{ \mathbf{h}_i^{(\ell)} + \phi_h^{(\ell)} \left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)} \right] \right\}. \tag{37}$$

In addition to updating node features, the EGNN refines residue coordinates using pairwise relative vectors. A bounded scalar coordinate-update coefficient is generated from the edge message, and the coordinate update is

$$\begin{aligned} \alpha_{ij}^{(\ell)} &= \tanh\left[\phi_x^{(\ell)} \left(\mathbf{m}_{ij}^{(\ell)} \right) \right] w_{ij}^{(\ell)}, \\ \mathbf{x}_i^{(\ell+1)} &= \mathbf{x}_i^{(\ell)} + 0.05 \, \frac{\sum_j \alpha_{ij}^{(\ell)} \left(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)} \right)}{\max\left(\sum_j w_{ij}^{(\ell)}, \, 1 \right)}. \end{aligned} \tag{38}$$

The coordinates are recentered after every layer to reduce numerical drift. Because the update depends only on scalar messages, distances, and relative coordinate vectors, the operation remains equivariant to rigid rotations and translations.

Three EGNN layers are stacked to progressively enlarge the spatial receptive field. The first layer primarily encodes direct geometric neighbors, whereas subsequent layers integrate higher-order spatial environments through repeated message passing. Unlike approaches that simply append pLDDT as an ordinary node feature, GHXTox uses pLDDT to directly modulate the strength of geometric messages, allowing structural uncertainty to influence representation learning at the propagation stage.

Continuous down-weighting does not imply that a low-pLDDT residue is biologically unimportant. Low confidence may arise from prediction error or potential conformational flexibility; the model reduces the reliability assigned to uncertain coordinate evidence while preserving the residue's sequence-semantic and physicochemical features.

2.4.4 Full-Atom Chemical-Site Message Passing

A Cα-based residue graph describes the global fold and spatial neighborhoods but cannot distinguish chemically distinct regions within the same side chain. For example, tyrosine contains both an aromatic ring and a hydroxyl group, whereas tryptophan contains an aromatic system and an indole nitrogen. Compressing each side chain into a single centroid would mix these chemically different moieties. GHXTox therefore extracts chemistry-aware pseudo-sites from the predicted full-atom PDB structure and processes them with a dedicated local chemical branch.

Up to two chemical sites are retained for each residue. Site identity is represented as an eight-dimensional multi-hot vector allowing overlapping types:

$$\mathbf{t}_{ir} \in \{0,1\}^8, \tag{39}$$

corresponding to positive, negative, donor, acceptor, aromatic, hydrophobic, sulfur, and conformational, where r indexes the chemical site within residue i. Multiple types may be assigned to the same site. For example, the cysteine sulfur site is labeled as donor, hydrophobic, and sulfur-containing, whereas the proline ring is labeled as hydrophobic and conformationally constrained. The aromatic ring and hydroxyl group of tyrosine are represented as separate sites, as are the aromatic system and the indole NE1 nitrogen (donor) of tryptophan.

A site coordinate is obtained from the mean position of the heavy atoms defining the corresponding chemical moiety. Site orientation is represented by either a bond direction or an unsigned plane normal. Because the sign of a plane normal is arbitrary, downstream directional features use absolute cosines, preserving rotation invariance and eliminating orientation-sign ambiguity.

The initial representation of site r in residue i is

$$\mathbf{z}_{ir}^{(0)} = \phi_{\text{site}}\left[\mathbf{t}_{ir} \parallel q_i \parallel o_{ir}^{\text{valid}}\right], \tag{40}$$

where q_i is the pLDDT score of the parent residue and o_{ir}^{valid} indicates whether a reliable orientation could be constructed; the input is projected into a 64-dimensional hidden space.

Chemical-site edges are established between valid sites separated by no more than 16 Å. In the current configuration, same-residue site edges are retained, preserving spatial relationships between chemically different moieties in complex side chains. Each edge contains four complementary feature families. First, the raw distance in Å is encoded using 16 radial basis functions, $\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \in \mathbb{R}^{16}$. Second, a peptide-size-normalized distance is constructed. The radius of gyration of the Cα coordinates is

$$R_g = \sqrt{\frac{1}{L}\sum_{i=1}^{L} \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \qquad \bar{\mathbf{x}} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i, \qquad \bar{d}_{uv} = \frac{d_{uv}}{R_g}, \tag{41}$$

where $\bar{\mathbf{x}}$ denotes the centroid of the valid Cα coordinates, and the normalized distance is encoded by eight radial basis functions over the interval 0–4, $\boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \in \mathbb{R}^8$. The raw representation

describes the physical separation in Å, whereas the normalized representation describes the separation relative to the overall peptide size. Their combination allows the model to distinguish, for example, an 8 Å contact in a compact short peptide from an 8 Å contact in a larger peptide.

Third, four rotation-invariant orientation features are used:

$$\mathbf{o}_{uv} = \left[|\mathbf{o}_u \cdot \mathbf{r}_{uv}|,\ |\mathbf{o}_v \cdot \mathbf{r}_{vu}|,\ |\mathbf{o}_u \cdot \mathbf{o}_v|,\ o_{uv}^{\text{valid}} \right], \tag{42}$$

which describe the alignment of the source orientation with the intersite direction, the alignment of the target orientation with the reverse direction, the mutual orientation of the two sites, and orientation availability. Fourth, six explicit compatibility channels are defined:

$$\mathbf{c}_{uv}^{\text{chem}} = [c_{\text{opp}}, c_{\text{DA}}, c_{\text{aro}}, c_{\text{cat-}\pi}, c_{\text{SS}}, c_{\text{hyd}}], \tag{43}$$

representing opposite-charge, donor–acceptor, aromatic–aromatic, positive–aromatic (cation–π), sulfur–sulfur, and hydrophobic–hydrophobic compatibility. These channels encode candidate chemical compatibility patterns available to the model; they do not imply that a true salt bridge, hydrogen bond, cation–π interaction, disulfide bond, or hydrophobic core is present in the predicted structure.

The full chemical-site edge representation is

$$\mathbf{e}_{uv}^{\text{chem}} = \left[\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \parallel \boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \parallel \mathbf{o}_{uv} \parallel \mathbf{c}_{uv}^{\text{chem}} \parallel \sqrt{q_u q_v} \right]. \tag{44}$$

At layer l, the site message, its pairwise-confidence-weighted form, and the confidence-normalized aggregation are

$$\begin{aligned} \mathbf{m}_{uv}^{(l)} &= \phi_{\text{chem}}^{(l)} \left[\mathbf{z}_u^{(l)} \parallel \mathbf{z}_v^{(l)} \parallel \mathbf{e}_{uv}^{\text{chem}} \right], \\ \tilde{\mathbf{m}}_{uv}^{(l)} &= \sqrt{q_u q_v}\, \mathbf{m}_{uv}^{(l)}, \\ \bar{\mathbf{m}}_v^{(l)} &= \frac{\sum_{u \in \mathcal{N}(v)} \tilde{\mathbf{m}}_{uv}^{(l)}}{\max\left(\sum_{u \in \mathcal{N}(v)} \sqrt{q_u q_v},\ 1 \right)}. \end{aligned} \tag{45}$$

After two message-passing layers, valid sites belonging to the same residue are mean-pooled, and a residue-level chemical residual is obtained:

$$\begin{aligned} \Delta \mathbf{h}_i^{\text{chem}} &= \mathbf{W}_{\text{chem}} \operatorname*{Mean}_{r \in \mathcal{S}_i} \left(\mathbf{z}_{ir}^{(2)} \right), \\ \mathbf{W}_{\text{chem}}^{(0)} &= 0, \quad \mathbf{b}_{\text{chem}}^{(0)} = 0, \\ \mathbf{h}_i^{\text{sp}} &= \mathbf{h}_i^{\text{EGNN}} + \Delta \mathbf{h}_i^{\text{chem}}, \end{aligned} \tag{46}$$

where the projection $\mathbf{W}_{\text{chem}}$ and its bias $\mathbf{b}_{\text{chem}}$ are initialized to zero, so that $\Delta \mathbf{h}_i^{\text{chem}} = 0$ at the beginning of training. This conservative residual design allows the new chemical-site branch to progressively learn useful corrections from an intact geometric backbone rather than perturbing the existing spatial representation with a randomly initialized contribution.

2.4.5 Availability-Masked Conformer-Ensemble Residual

A single predicted structure provides a deterministic spatial anchor, whereas a peptide may adopt multiple related conformations. Running the complete geometric network independently on every conformer would be computationally expensive and could allow the model to exploit differences between conformer generators or uneven ensemble coverage. The current GHXTox model therefore uses a lightweight conformer-statistics residual. The single-structure expert remains the base predictor, and ensemble statistics are permitted to correct its logit only when valid conformer information is available.

For a peptide with M aligned conformers, the residue-level root-mean-square fluctuation is

$$\mathrm{RMSF}_i = \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left\|\mathbf{x}_i^{(m)} - \bar{\mathbf{x}}_i\right\|_2^2}, \qquad \bar{\mathbf{x}}_i = \frac{1}{M}\sum_{m=1}^{M}\mathbf{x}_i^{(m)}. \tag{47}$$

For each residue pair i, j, the mean distance and the distance standard deviation are

$$\bar{d}_{ij} = \frac{1}{M}\sum_{m=1}^{M} d_{ij}^{(m)}, \qquad \sigma_{ij}^{d} = \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left(d_{ij}^{(m)} - \bar{d}_{ij}\right)^2}. \tag{48}$$

For a distance cutoff $r \in \{6, 8, 10\}$ Å, the contact occupancy and the 8 Å contact-occupancy uncertainty are

$$O_{ij}^{(r)} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\left(d_{ij}^{(m)} < r\right), \qquad U_{ij}^{(8)} = O_{ij}^{(8)}\left(1 - O_{ij}^{(8)}\right). \tag{49}$$

Values close to zero or one indicate contacts that are consistently absent or present across the sampled conformers, whereas values near 0.5 indicate contact switching.

The ensemble statistics are organized into an eight-dimensional residue-level descriptor $\mathbf{r}_i^{\mathrm{conf}} \in \mathbb{R}^8$, containing the residue RMSF; the mean pairwise distance to other residues; the standard deviation of the mean pairwise distances associated with the residue; the mean pairwise-distance standard deviation; the mean contact occupancies at 6, 8, and 10 Å; and the mean 8 Å contact-occupancy uncertainty, which together form the eight-dimensional descriptor. A twelve-dimensional peptide-level descriptor $\mathbf{g}^{\mathrm{conf}} \in \mathbb{R}^{12}$ is also constructed, containing the normalized number of conformers; the mean, standard deviation, and maximum of RMSF; the mean, standard deviation, and maximum of the conformer RMSD; the mean and standard deviation of the radius of gyration; the mean and maximum of the pair-distance standard deviation; and the global mean of $U_{ij}^{(8)}$, which together form the twelve-dimensional descriptor. Before neural encoding, distance- and fluctuation-based statistics are rescaled by fixed constants to place heterogeneous descriptors on comparable numerical ranges; the exact scaling factors are reported in the implementation-details table.

The residue-level ensemble features are encoded and aggregated using both masked mean and masked maximum pooling, whereas the global descriptor is processed by a separate encoder:

$$\mathbf{h}_{\mathrm{conf}}^{\mathrm{res}} = \left[\mathrm{MeanPool}\left(\phi_r\left(\mathbf{R}^{\mathrm{conf}}\right)\right) \,\|\, \mathrm{MaxPool}\left(\phi_r\left(\mathbf{R}^{\mathrm{conf}}\right)\right)\right], \qquad \mathbf{h}_{\mathrm{conf}}^{\mathrm{global}} = \phi_g\left(\mathbf{g}^{\mathrm{conf}}\right). \tag{50}$$

The base structural embedding, the residue-level ensemble embedding, and the global ensemble embedding are concatenated to produce a raw logit residual:

$$\Delta z_{\mathrm{conf}}^{\mathrm{raw}} = \phi_{\mathrm{conf}}\left[\mathbf{h}_{\mathrm{base}} \,\|\, \mathbf{h}_{\mathrm{conf}}^{\mathrm{res}} \,\|\, \mathbf{h}_{\mathrm{conf}}^{\mathrm{global}}\right]. \tag{51}$$

The final linear layer of the conformer residual head is initialized to zero. Using the availability indicator c_{conf} defined in Eq. (27), the learnable conformer weight, the effective residual, and the final structural-expert logit are

$$\begin{aligned} \beta_{\mathrm{conf}}^{(0)} &= \mathrm{logit}(0.25), \\ \lambda_{\mathrm{conf}} &= \sigma\left(\beta_{\mathrm{conf}}\right), \\ \Delta z_{\mathrm{conf}} &= c_{\mathrm{conf}}\,\lambda_{\mathrm{conf}}\,\Delta z_{\mathrm{conf}}^{\mathrm{raw}}, \\ z_{\mathrm{SE3D}} &= z_{\mathrm{base}} + \Delta z_{\mathrm{conf}}, \qquad c_{\mathrm{conf}} = 0 \;\Rightarrow\; z_{\mathrm{SE3D}} = z_{\mathrm{base}}. \end{aligned} \tag{52}$$

where the effective conformer weight is initialized to 0.25 at the beginning of training through $\beta_{\mathrm{conf}}^{(0)} = \mathrm{logit}(0.25)$, so that $\sigma(\beta_{\mathrm{conf}}^{(0)}) = 0.25$. Zero-filled feature tensors alone would not guarantee a zero correction because the residual network may contain nonzero biases; the additional hard availability mask ensures an exact fallback to the single-structure prediction. The conformer branch is permitted only to refine the single-structure predictor rather than replace the geometric backbone, reducing the risk of overfitting to limited and uneven ensemble coverage.

2.4.6 pLDDT-Aware Sequence–Structure Cross-Attention

After the EGNN and the chemical-site residual are applied, the model obtains the chemical-enhanced spatial representation

$$\mathbf{H}^{\mathrm{sp}} = [\mathbf{h}_1^{\mathrm{sp}}, \ldots, \mathbf{h}_L^{\mathrm{sp}}]^\top \in \mathbb{R}^{L\times 128}, \tag{53}$$

where $\mathbf{h}_i^{\mathrm{sp}}$ is defined in Eq. (46). Spatial information alone does not determine whether a structural pattern is relevant to the current sequence context. For example, similar hydrophobic contacts may have different implications depending on residue identities and sequence positions. GHXTox therefore uses pLDDT-aware cross-attention within the structural expert, allowing contextualized sequence states to selectively retrieve spatial evidence.

A node-level gate is first constructed for each residue. The base gate uses the 42-dimensional residue descriptors and the pLDDT score:

$$\begin{aligned} g_i^{\mathrm{base}} &= \phi_{\mathrm{node}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel q_i\right], \\ \mathbf{c}_i^{\mathrm{node}} &= \left[\frac{q_i-\mu_q}{T_q},\ \frac{(q_i-\mu_q)^2}{T_q^2},\ \mathbb{I}(q_i<0.55)\right], \\ \mu_q &= 0.70, \quad T_q = 0.12, \end{aligned} \tag{54}$$

where the residual confidence vector $\mathbf{c}_i^{\mathrm{node}}$ increases the sensitivity of the gate to the distribution of structural confidence. The residual scale and the final node gate are

$$r_i^{\mathrm{node}} = 0.5 + \sigma\left(\mathbf{w}_n^\top \mathbf{c}_i^{\mathrm{node}} + b_n\right), \qquad g_i^{\mathrm{node}} = \mathrm{clamp}\left(g_i^{\mathrm{base}}\, r_i^{\mathrm{node}},\ 0, 1\right), \tag{55}$$

and the spatial representation is continuously down-weighted as

$$\widehat{\mathbf{H}}^{\mathrm{sp}} = \mathbf{H}^{\mathrm{sp}} \odot \mathbf{g}^{\mathrm{node}}, \qquad \mathbf{g}^{\mathrm{node}} = \left[g_1^{\mathrm{node}}, \ldots, g_L^{\mathrm{node}}\right]^\top \in [0,1]^L, \tag{56}$$

where the node-level gate vector $\mathbf{g}^{\mathrm{node}}$ is broadcast across the feature dimension and each gate value is clamped to $[0, 1]$.

In the cross-attention operation, the contextualized sequence representation serves as the query, and the gated spatial representation serves as the key and value:

$$\mathbf{H}^{\mathrm{cross}} = \mathrm{MHA}\left(Q = \mathbf{H}^{\mathrm{seq}}, K = \widehat{\mathbf{H}}^{\mathrm{sp}}, V = \widehat{\mathbf{H}}^{\mathrm{sp}}\right). \tag{57}$$

This direction has a direct interpretation: each residue position retrieves spatial evidence according to its contextual sequence representation. The operation is not a direct feature interaction between the LoRA-ProtT5–Mamba expert and the structural expert; it occurs internally between the ESM2-enhanced sequence states and the geometric states of the structural expert.

In addition to residue-level gating, a peptide-level gate is generated from the overall structure-confidence profile. The model first computes

$$\bar{q} = \frac{1}{L}\sum_{i=1}^{L} q_i, \qquad q_{\min} = \min_i q_i, \qquad r_{\mathrm{low}} = \frac{1}{L}\sum_{i=1}^{L} \mathbb{I}(q_i < 0.55), \tag{58}$$

and constructs the peptide-level confidence vector $\mathbf{c}^{\mathrm{pep}} = [(\bar{q}-\mu_q)/T_q,\ (q_{\min}-\mu_q)/T_q,\ r_{\mathrm{low}}]$. The base peptide-level gate and its residual-adjusted form are

$$\begin{aligned} g_{\mathrm{pep}}^{\mathrm{base}} &= \phi_{\mathrm{global}}\left[\bar{q},\ q_{\min},\ r_{\mathrm{low}}\right], \\ r_{\mathrm{pep}} &= 0.5 + \sigma\left(\mathbf{w}_{\mathrm{pep}}^\top \mathbf{c}^{\mathrm{pep}} + b_{\mathrm{pep}}\right), \\ g_{\mathrm{pep}} &= \mathrm{clamp}\left(g_{\mathrm{pep}}^{\mathrm{base}}\, r_{\mathrm{pep}},\ 0, 1\right), \end{aligned} \tag{59}$$

where the peptide-level residual scale is analogous to, but separately parameterized from, the node-level residual scale, and $g_{\text{pep}} \in [0, 1]$ denotes the scalar peptide-level structural gate.

The final residue-level fusion representation is

$$\mathbf{H}^{\text{fused}} = \text{LN}\left[\mathbf{H}^{\text{seq}} \parallel g_{\text{pep}}\, \mathbf{H}^{\text{cross}}\right]. \tag{60}$$

The two gating levels answer complementary questions: how reliable the structural evidence is at an individual residue position, and how strongly the structural branch should be trusted for the peptide as a whole. The sequence representation is always retained, whereas the structural context is dynamically scaled according to prediction confidence. Consequently, when the predicted structure has low overall confidence, the structural expert can still rely on its internal sequence-semantic pathway without allowing uncertain coordinates to dominate the classification.

2.4.7 Global Feature Integration and Structural Classification

After pLDDT-aware cross-attention, each valid residue is represented by a 256-dimensional fused vector obtained by concatenating the 128-dimensional internal sequence state with the 128-dimensional structural context:

$$\mathbf{H}^{\text{fused}} \in \mathbb{R}^{L \times 2d_h}, \qquad d_h = 128, \tag{61}$$

so that $\mathbf{H}^{\text{fused}} \in \mathbb{R}^{L \times 256}$. Masked mean pooling is applied over the valid residue positions:

$$\mathbf{h}_{\text{pool}} = \frac{\sum_{i=1}^{L} m_i\, \mathbf{H}_i^{\text{fused}}}{\sum_{i=1}^{L} m_i} \in \mathbb{R}^{256}, \tag{62}$$

where the denominator equals the number of valid residue positions and the mask excludes padding positions from the peptide representation. In contrast to maximum pooling, mean pooling integrates structural evidence distributed across multiple residue positions, which is appropriate when toxicity-related spatial patterns are not confined to a single local region.

The residue graph primarily models local and intermediate-range environments, whereas a finite number of message-passing layers may not explicitly recover all peptide-level physicochemical statistics. GHXTox therefore includes five global physicochemical descriptors, normalized to comparable scales before concatenation:

$$\mathbf{f}_{\text{pep}}^{\text{phys}} = \left[\frac{L}{128},\ \frac{Q_{\text{net}}}{L},\ f_{\text{aro}},\ f_{\text{Cys}},\ \frac{\bar{h}_{\text{hyd}}}{4.5}\right] \in \mathbb{R}^5, \tag{63}$$

representing the normalized peptide length, the length-normalized net charge, the aromatic-residue fraction, the cysteine fraction, and the scaled mean hydropathy. The classifier input is

$$\mathbf{h}_{\text{cls}} = \left[\mathbf{h}_{\text{pool}} \parallel \mathbf{f}_{\text{pep}}^{\text{phys}}\right] \in \mathbb{R}^{261}. \tag{64}$$

The base structural logit is produced by a three-layer multilayer perceptron, denoted compactly as

$$z_{\text{base}} = \phi_{\text{cls}}\left(\mathbf{h}_{\text{cls}}\right), \tag{65}$$

with dimensions $261 \to 128 \to 64 \to 1$, where each hidden layer applies GELU activation and dropout.

Applying the availability-masked conformer correction defined in Eq. (52) yields the final structural-expert logit, and the structural-expert probability is

$$z_{\text{SE3D}} = z_{\text{base}} + c_{\text{conf}}\, \lambda_{\text{conf}}\, \Delta z_{\text{conf}}^{\text{raw}}, \qquad p_{\text{SE3D}} = \sigma\left(z_{\text{SE3D}}\right),$$

as defined in Eq. (52). No standalone structural threshold is used to generate the final GHXTox class at this stage. Instead, p_{SE3D} is passed directly to the top-level dual-expert fusion described in Section 2.2.3, where it is combined with the calibrated 1D expert probability p_{1D} using frozen fusion weights. A standalone structural threshold is used only for structural-branch ablations or independent expert evaluation and does not determine the final GHXTox prediction.

The structural expert can therefore be summarized as

$$\begin{aligned}&\text{sequence-enhanced residue semantics} + \text{confidence-aware geometry}\\ &+ \text{full-atom chemical sites} + \text{global physicochemical descriptors}\\ &+ \text{availability-masked conformer statistics} \longrightarrow p_{\mathrm{SE3D}}.\end{aligned}$$

The central contribution is not merely the inclusion of additional structural features, but the explicit separation of structural information by spatial scale and reliability. The geometric graph captures global residue relationships, the chemical-site branch resolves local functional moieties, the conformer residual summarizes sampled structural variation, and pLDDT continuously modulates edge propagation, chemical messages, and sequence–structure fusion. This design allows the structural expert to incorporate complementary 3D evidence while handling structure-prediction uncertainty in an explicit and auditable manner.

Overall, the structural expert integrates residue-level sequence semantics, confidence-aware geometry, full-atom chemical-site interactions, peptide-level physicochemical properties, and availability-masked conformer statistics. Rather than treating all predicted structural evidence as equally reliable, GHXTox continuously modulates its contribution using residue-level and peptide-level confidence signals. The following section describes the training, out-of-fold model-selection, calibration, and parameter-freezing protocols used to optimize the complete framework.

2.5 Training and Model-Selection Protocol

GHXTox consists of a task-adaptive 1D sequence expert and a sequence-enhanced 3D structural expert. Because the two experts differ substantially in input representation, parameter scale, and optimization behavior, they are not jointly optimized through a single end-to-end objective. Instead, expert-level representations are learned independently, after which probability calibration and top-level fusion are performed using training out-of-fold (OOF) predictions. This strategy allows each expert to be optimized under modality-appropriate settings while separating representation learning, checkpoint selection, probability calibration, fusion-parameter determination, and benchmark evaluation into clearly defined stages.

The training protocol follows three central principles. First, identical or highly similar peptides must not appear simultaneously in a training partition and the corresponding validation partition. Second, every prediction used for calibration, fusion-weight selection, or threshold determination must be generated by a model that was not trained on the corresponding sample. Third, Test1 and Test2 are not used for checkpoint selection, calibrator fitting, fusion search, or decision-threshold optimization in the current pipeline. Their prior observation during project development is disclosed, and they are therefore treated as frozen retrospective benchmarks rather than newly blinded external validation sets.

2.5.1 Group-Aware Five-Fold OOF Generation

The training set contains $N = 6387$ peptides. To reduce performance inflation caused by highly similar sequences, a fixed five-fold manifest is constructed by clustering the training sequences into homology groups. The fold-level homology groups were constructed through a separate sequence-clustering procedure, distinct from the dataset-level redundancy removal described in Section 2.1. The first operation determines which samples remain in the benchmark

dataset, whereas the second determines which retained samples must remain in the same cross-validation fold. Let

$$\mathcal{D} = \{(\mathcal{S}_i, y_i, g_i)\}_{i=1}^{N}, \qquad N = 6387, \tag{66}$$

denote the training data, where $\mathcal{S}_i$ is the peptide sequence, $y_i \in \{0, 1\}$ is its toxicity label, and g_i denotes its sequence-homology group. The data are partitioned into five disjoint validation folds:

$$\begin{aligned} \mathcal{D} &= \bigcup_{k=1}^{5} \mathcal{D}_k, \qquad\qquad \mathcal{D}_k \cap \mathcal{D}_l = \varnothing, \quad k \neq l, \\ g_i = g_j &\Longrightarrow f(i) = f(j), \end{aligned} \tag{67}$$

where $f(i)$ denotes the fold assigned to sample i. All peptides belonging to the same homology group are therefore placed in the same fold, preventing closely related sequences from crossing the training–validation boundary.

For fold k, the model is trained only on the complement $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$ and evaluated on the unseen fold $\mathcal{D}_k$. The OOF probability of sample $i \in \mathcal{D}_k$ is defined as

$$\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k, \qquad \widehat{p}_i^{\,\mathrm{OOF}} = F_{\boldsymbol{\theta}^{(-k)}}\left(\mathcal{S}_i\right), \quad i \in \mathcal{D}_k, \tag{68}$$

where $\boldsymbol{\theta}^{(-k)}$ denotes the parameters learned exclusively from $\mathcal{D}_{-k}$. Each training peptide consequently receives exactly one probability from a model that did not observe that peptide during optimization.

For the 1D expert, full seed-specific OOF predictions are generated for the training set, and the three probability vectors are aligned by sample index and averaged:

$$\overline{p}_{i,\mathrm{1D}}^{\,\mathrm{OOF}} = \frac{1}{3} \sum_{s \in \{42,123,2025\}} \widehat{p}_{i,\mathrm{1D}}^{\,(s,\mathrm{OOF})}. \tag{69}$$

The resulting OOF ensemble scores are used to fit the Platt calibrator described in Section 2.3.4.

The structural expert uses the same fixed group-aware folds. Each fold-specific checkpoint produces probabilities only for its held-out validation fold:

$$\widehat{p}_{i,\mathrm{SE3D}}^{\,\mathrm{OOF}} = G_{\boldsymbol{\phi}^{(-k)}}\left(\mathcal{S}_i, \mathcal{X}_i, \mathcal{A}_i, \mathbf{q}_i, \mathcal{C}_i\right), \qquad i \in \mathcal{D}_k, \tag{70}$$

where $\mathcal{X}_i$, $\mathcal{A}_i$, $\mathbf{q}_i$, and $\mathcal{C}_i$ denote the residue geometry, full-atom chemical information, pLDDT confidence, and available conformer-ensemble statistics, respectively; $\mathcal{C}_i$ may be unavailable and is handled by the conformer-availability mask defined in Section 2.4.1.

Before combining the expert-level OOF predictions, the implementation verifies immutable sample indices (`source_index`), peptide sequences, toxicity labels, and fold assignments. Duplicate sample indices or mismatches in sequence, label, or fold assignment cause the fusion procedure to terminate rather than silently aligning prediction files by row order. OOF predictions are treated as **training-only model-selection assets**. Although OOF predictions are generated by models that did not observe the corresponding validation samples, parameters selected from the complete OOF table remain development-stage choices and must be evaluated on independent test sets.

2.5.2 Loss Functions and Class-Imbalance Handling

The 1D and 3D experts are optimized using separate objectives. This prevents uneven structural coverage, conformer availability, and confidence variation from directly perturbing the language-model optimization, and it avoids the large ProtT5 parameter space from dominating the smaller structure-specific residual modules.

Loss for the 1D expert. As described in Section 2.3.3, the sequence expert is trained using positive-class-weighted binary cross-entropy:

$$\mathcal{L}_{\mathrm{1D}} = -\frac{1}{B}\sum_{i=1}^{B}\left[w_{+}y_i \log \sigma(z_i) + (1-y_i)\log\left(1-\sigma(z_i)\right)\right], \qquad w_{+} = \frac{N_{-}}{N_{+}}, \tag{71}$$

where the class counts N_{+} and N_{-} are independently calculated from each fold's training partition. The current 1D training pipeline does not use Borderline-SMOTE or sequence point-mutation augmentation. The weighted loss increases the contribution of the minority toxic class without introducing synthetic near-duplicate peptides.

Base geometric and chemical-site stage. The structural expert is optimized in stages. The full-atom chemical-site branch is first initialized from a previously trained pLDDT-aware geometric model, while the remaining backbone is frozen. This stage uses positive-class-weighted BCE:

$$\mathcal{L}_{\mathrm{chem}} = -\frac{1}{B}\sum_{i=1}^{B}\left[w_{+}y_i \log \sigma\left(z_i^{\mathrm{base}}\right) + (1-y_i)\log\left(1-\sigma\left(z_i^{\mathrm{base}}\right)\right)\right], \tag{72}$$

where w_{+} is defined as in Section 2.3.3 and is recalculated from each fold's training partition.

During training, pLDDT-dependent Gaussian coordinate perturbations are applied to reduce sensitivity to local errors in predicted structures. The perturbation scale for residue i is

$$\sigma_i = \sigma_0\left[1+\kappa\left(1-q_i\right)\right], \qquad \sigma_0 = 0.12\ \text{Å}, \quad \kappa = 2.0, \tag{73}$$

and residue-wise perturbations are drawn as

$$\boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \sigma_i^2\mathbf{I}_3\right), \qquad \widetilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\varepsilon}_i. \tag{74}$$

The same residue-wise displacement is applied consistently to the corresponding backbone, functional-group, and chemical-site coordinates, thereby preserving their intra-residue relative geometry. No coordinate perturbation is applied during validation or inference. Low-pLDDT residues therefore receive stronger perturbations than high-confidence residues. This augmentation uses pLDDT to modulate the perturbation magnitude but does not treat pLDDT as an exact estimate of coordinate-error variance.

Conformer-ensemble residual stage. During conformer-residual training, the pLDDT-aware geometric backbone, internal sequence encoder, chemical-site branch, and base classifier are frozen. Only the residue-level conformer encoder, the global conformer encoder, the conformer-residual MLP, and the conformer logit-residual parameters are optimized. This stage is initialized from the trained chemical-site checkpoint. To avoid applying a second class-frequency correction on top of the existing base classifier, the current conformer-residual configuration uses unweighted BCE and disables coordinate perturbation, with a residual-magnitude penalty:

$$\mathcal{L}_{\mathrm{conf}} = \mathrm{BCEWithLogits}\left(z_{\mathrm{SE3D}}, y\right) + \lambda_{\Delta}\frac{\sum_{i=1}^{B} c_i\left|\Delta z_{\mathrm{conf},i}\right|}{\max\left(\sum_{i=1}^{B} c_i,\ 1\right)}, \qquad \lambda_{\Delta} = 0.005, \tag{75}$$

where the residual-magnitude term is computed only on samples with available conformer information. It limits large logit corrections and encourages the conformer branch to act as a conservative refinement rather than re-learning the complete toxicity decision boundary.

The final conformer-residual layer is zero-initialized, and its output is additionally controlled by the hard `conformer_available` mask. Zero initialization makes the initial model equivalent to the single-structure expert, whereas the hard mask guarantees a zero residual for samples without conformer evidence. Together, these mechanisms provide a stable incremental extension of the structural model.

2.5.3 Optimization, Checkpoint Selection, and Hyperparameter Settings

Both experts are optimized using AdamW [28], but different learning rates, training durations, and checkpoint-selection criteria are employed. All random-number generators are initialized with fixed seeds. Data are shuffled only during training, whereas validation and inference loaders preserve deterministic ordering. Table 2 summarizes the optimization and model-selection settings of both experts.

Table 2: Main optimization and model-selection settings of the two GHXTox experts.

Module or stage	Batch size	Learning rate	Weight decay	Epochs	Checkpoint rule
1D LoRA-ProtT5 + Bi-Mamba	16	3×10^{-4}	1×10^{-2}	Fixed 15	No early stopping
3D chemical-site refinement	32	5×10^{-4}	1×10^{-4}	Max. 30	Best validation MCC; patience 8
Conformer residual refinement	32	1×10^{-3}	5×10^{-4}	Max. 50	Minimum validation loss; patience 8
Single-structure chemical-site backbone	32	Stage-specific	Stage-specific	Fixed 8	Median fold-selected epoch
PepFlow pilot250 conformer residual	32	Stage-specific	Stage-specific	Fixed 12	Median fold-selected epoch

1D expert. The LoRA-ProtT5 and bidirectional Mamba expert is trained with a batch size of 16, a learning rate of 3×10^{-4}, a weight decay of 1×10^{-2}, a fixed duration of 15 epochs, dropout of 0.1, FP16 mixed precision, and ProtT5 gradient checkpointing, using random seeds 42, 123, and 2025. This configuration yields approximately 13.1 million trainable parameters, comprising the LoRA updates and the downstream sequence-classification modules, whereas the pretrained base transformations and embeddings remain frozen. Every fold model and every full-data model is trained for a fixed 15 epochs. Validation-based early stopping and a separate learning-rate scheduler are not used. A fixed training budget ensures consistency across folds and random seeds and avoids differences in training duration caused by noisy fold-level validation metrics. After cross-validation, three models are independently trained on the complete training set using the same configuration. The deployed sequence probability is obtained by averaging these three full-data models rather than selecting the seed with the highest validation performance.

3D chemical-site stage. The full-atom chemical-site refinement uses a batch size of 32, a learning rate of 5×10^{-4}, a weight decay of 1×10^{-4}, a maximum duration of 30 epochs, early-stopping patience of 8 epochs, base structural dropout of 0.22, and chemical-site dropout of 0.1. This stage is initialized from a pLDDT-aware geometric checkpoint, and only the chemical-site branch is made trainable. Training is terminated when the validation MCC does not improve for eight consecutive epochs, and the checkpoint with the highest validation MCC is retained.

Conformer-residual stage. The conformer-ensemble residual stage uses a batch size of 32, a learning rate of 1×10^{-3}, a weight decay of 5×10^{-4}, a maximum duration of 50 epochs, early-stopping patience of 8 epochs, conformer-residual dropout of 0.25, a conformer hidden dimension of 32, an initial conformer weight of 0.25, and a residual L1 coefficient of 0.005. Validation loss is used instead of training accuracy to select the conformer checkpoint, reducing sensitivity to threshold fluctuations arising from limited conformer coverage. Only conformer-related modules are optimized, whereas the geometric and chemical backbone remains frozen.

During structural-expert training, gradients of all trainable parameters are clipped to a maximum norm of 5.0 ($\|\nabla_\phi \mathcal{L}\|_2 \leq 5.0$), limiting unusually large gradients caused by local residual corrections or coordinate perturbations.

When a full-data structural checkpoint is required, the number of refitting epochs is not selected using either benchmark test set. Instead, a fixed epoch count is obtained from the median of the fold-specific selected epochs. For the single-structure chemical-site backbone, the

recorded fold-level epochs are 8, 8, 14, 2, and 4, whose median is 8, and the full-data refit is trained for eight fixed epochs. The conformer-statistics residual is a separate stage trained only on the PepFlow pilot250 subset; its recorded fold-level epochs are 23, 18, 9, 12, and 1, whose median is 12, and the full pilot250 refit is trained for twelve fixed epochs.

2.5.4 OOF-Based Calibration, Fusion-Weight Selection, and Threshold Freezing

After the expert-level OOF predictions have been generated, model selection proceeds at the probability level. No ProtT5, Mamba, EGNN, chemical-site, or conformer-residual parameters are updated during this stage. Only the training OOF probabilities are used to determine the calibrator, fusion weight, and final decision threshold.

The three seed-specific 1D OOF probabilities are first averaged and calibrated using the monotonic Platt mapping described in Section 2.3.4:

$$p_{i,\mathrm{1D}}^{\mathrm{cal}} = \sigma\left[a\,\mathrm{logit}\left(\bar{p}_{i,\mathrm{1D}}^{\mathrm{OOF}}\right) + b\right], \qquad a = \exp(\eta) > 0, \tag{76}$$

where the scale is parameterized as a positive value, $a = \exp(\eta) > 0$, as described in Section 2.3.4. Platt calibration is applied only to the 1D ensemble. The structural-expert probability is not subjected to an additional top-level calibrator, and no new calibrator is fitted after the final fusion.

The calibrated sequence probability and the structural-expert probability are combined in probability space:

$$p_i(w) = w\,p_{i,\mathrm{1D}}^{\mathrm{cal}} + (1-w)\,p_{i,\mathrm{SE3D}}^{\mathrm{OOF}}, \qquad 0 \le w \le 1, \tag{77}$$

where w denotes the 1D expert weight and $1-w$ denotes the sequence-enhanced structural-expert weight. No additional stacking classifier or sample-specific routing network is fitted at this stage, reducing the risk of second-level overfitting to the finite OOF table.

For this top-level fusion, $p_{i,\mathrm{SE3D}}^{\mathrm{OOF}}$ is the pure `structure_probability` emitted by the pLDDT-aware structural expert. It is not the probability of an earlier sequence–structure fusion model and contains no contribution from the current or historical 1D predictor. This strict interface prevents duplicated sequence evidence and avoids mixing old and new model generations during OOF weight selection.

For a candidate weight w and threshold τ, the predicted class is

$$\hat{y}_i(w,\tau) = \mathbb{I}\left[p_i(w) \ge \tau\right], \tag{78}$$

and the frozen parameters are selected according to

$$(w^*, \tau^*) = \arg\max_{w,\tau}\ \mathrm{MCC}\left(\mathbf{y}, \hat{\mathbf{y}}(w,\tau)\right). \tag{79}$$

MCC is used as the primary objective because of the pronounced class imbalance in the benchmark data, particularly in Test2. Balanced accuracy was used as the secondary selection criterion when candidate operating points produced identical MCC values under the evaluated search procedure.

The frozen values are

$$w_{\mathrm{1D}}^* = 0.72, \qquad w_{\mathrm{SE3D}}^* = 0.28, \qquad \tau^* = 0.47022256531284556 \approx 0.4702, \tag{80}$$

where the full-precision threshold is rounded to 0.4702 when reported in the final decision rule. The final probability and class prediction are therefore

$$p_{\mathrm{GHXTox}} = 0.72\,p_{1D}^{\mathrm{cal}} + 0.28\,p_{\mathrm{SE3D}}, \qquad \hat{y} = \mathbb{I}\left[p_{\mathrm{GHXTox}} \ge 0.4702225653\right]. \tag{81}$$

A second-level leave-one-fold-out analysis is additionally performed as a stability diagnostic. In each iteration, w and τ are selected using four OOF folds and applied to the remaining fold.

This analysis assesses whether the fusion parameters are disproportionately dependent on a particular fold. However, because every upstream prediction stream is not fully regenerated within a separately nested outer loop, this diagnostic is not described as strict nested external validation.

The final values of 0.72/0.28 and 0.47022256531284556 are frozen from the complete training OOF table. The corresponding fused OOF metrics describe training-stage model selection rather than benchmark test-set performance. After freezing, the same calibration parameters, fusion weights, and threshold are applied unchanged to Test1 and Test2:

$$(a^*, b^*, w^*, \tau^*)_{\text{Test}} = (a^*, b^*, 0.72, 0.47022256531284556)_{\text{Training OOF}}\,. \tag{82}$$

Neither Test1 nor Test2 labels are used for recalibration or parameter adjustment. The frozen parameters, input prediction files, fusion space, and decision threshold are stored in machine-readable artifacts to permit auditing of the final evaluation protocol.

2.6 Evaluation Metrics

To comprehensively evaluate GHXTox for peptide toxicity prediction, model performance is assessed from three complementary perspectives: classification performance at a fixed operating threshold, ranking ability based on continuous probability scores, and probabilistic calibration. Reliance on a single metric may conceal important differences in minority-class recognition, false-positive control, or probability reliability. We therefore jointly report confusion-matrix metrics, the area under the receiver operating characteristic curve (AUROC), the area under the precision–recall curve (AUPRC), the Brier score, and the expected calibration error (ECE).

Let the evaluation set contain N peptides. The ground-truth label of peptide i is

$$y_i \in \{0, 1\}, \qquad p_i = \mathbb{P}\left(y_i = 1 \mid \mathcal{S}_i\right), \tag{83}$$

where $y_i = 1$ denotes a toxic peptide, $y_i = 0$ denotes a non-toxic peptide, and $\mathcal{S}_i$ denotes the peptide sequence. The binary prediction is obtained using a fixed decision threshold τ:

$$\hat{y}_i = \mathbb{I}\left(p_i \geq \tau\right). \tag{84}$$

For the final GHXTox model, the threshold is frozen using the complete training OOF table, as described in Section 2.5.4. Its full-precision value is $\tau^* = 0.47022256531284556$, which is rounded to $\tau = 0.4702$ for presentation in the manuscript and result tables. The same frozen threshold is applied unchanged to Test1 and Test2, without dataset-specific retuning. Threshold-dependent metrics are calculated from $\hat{y}_i$, whereas AUROC, AUPRC, the Brier score, and ECE are calculated directly from the continuous probability p_i.

Because Test2 contains only 46 toxic peptides and 536 non-toxic peptides, its class distribution is strongly imbalanced. Accuracy alone may therefore provide an incomplete description of toxic-peptide recognition. Balanced accuracy (BACC) and the Matthews correlation coefficient (MCC) are emphasized as balanced summary measures; sensitivity (SN) and specificity (SP) are reported to expose the trade-off between missed toxic peptides and false alarms; and AUPRC and calibration metrics are reported together with AUROC.

2.6.1 Threshold-Dependent Classification Metrics

At a fixed decision threshold, the predictions can be summarized by four confusion-matrix counts:

$$\begin{aligned} \mathrm{TP} &= \sum_{i=1}^{N} \mathbb{I}\left(y_i = 1,\ \hat{y}_i = 1\right), \quad \mathrm{TN} = \sum_{i=1}^{N} \mathbb{I}\left(y_i = 0,\ \hat{y}_i = 0\right), \\ \mathrm{FP} &= \sum_{i=1}^{N} \mathbb{I}\left(y_i = 0,\ \hat{y}_i = 1\right), \quad \mathrm{FN} = \sum_{i=1}^{N} \mathbb{I}\left(y_i = 1,\ \hat{y}_i = 0\right), \end{aligned} \tag{85}$$

where TP denotes toxic peptides correctly identified as toxic, TN denotes non-toxic peptides correctly identified as non-toxic, FP denotes non-toxic peptides incorrectly predicted as toxic, and FN denotes toxic peptides incorrectly predicted as non-toxic.

Accuracy. Accuracy measures the proportion of correctly classified samples:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{86}$$

ACC provides an overall correctness measure but may be dominated by the majority class in strongly imbalanced datasets. A model biased toward non-toxic predictions may still achieve high ACC when non-toxic peptides greatly outnumber toxic peptides. ACC is therefore not used as the sole measure of predictive performance.

Sensitivity and specificity. Sensitivity, also referred to as recall or the true-positive rate, measures the fraction of toxic peptides correctly identified:

$$\mathrm{SN} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{SP} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}. \tag{87}$$

A high SN indicates that relatively few toxic peptides are missed. In a toxicity-screening context, FN corresponds to potentially hazardous peptides that are not detected, making SN an important measure of toxic-class coverage. Specificity, or the true-negative rate, measures the fraction of non-toxic peptides correctly rejected. A high SP indicates that relatively few non-toxic candidates are falsely labeled as toxic. SN and SP should be interpreted jointly: lowering the decision threshold generally increases SN but may increase FP and reduce SP, whereas increasing the threshold may produce the opposite trade-off.

Precision and F1 score. Precision measures the proportion of predicted toxic peptides that are truly toxic:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad F_1 = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{SN}}{\mathrm{Precision} + \mathrm{SN}} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{88}$$

Precision reflects the reliability of positive toxicity predictions. Because Precision depends on the prevalence of toxic peptides, comparisons across datasets with different class ratios should be interpreted together with the corresponding sample composition. The F1 score balances positive-prediction purity and toxic-peptide recall; however, it does not explicitly include TN and therefore does not fully describe performance on the non-toxic class.

Balanced accuracy. Balanced accuracy is the arithmetic mean of sensitivity and specificity:

$$\mathrm{BACC} = \frac{\mathrm{SN} + \mathrm{SP}}{2}. \tag{89}$$

BACC assigns equal importance to the toxic and non-toxic classes, preventing the larger class from directly dominating the final score. A model performing well on one class but poorly on the other will therefore have a noticeably lower BACC than ordinary ACC.

Matthews correlation coefficient. The Matthews correlation coefficient incorporates all four elements of the confusion matrix:

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}, \tag{90}$$

with $-1 \leq \mathrm{MCC} \leq 1$. An MCC of 1 indicates perfect agreement, 0 indicates no overall correlation between predictions and labels, and -1 indicates complete disagreement. Because MCC jointly accounts for TP, TN, FP, and FN, it was used as the primary summary metric for fixed-threshold classification and as the primary objective for OOF fusion-parameter selection [29].

Numerical safeguards were applied when metric denominators were zero. All threshold-dependent metrics were calculated using the same OOF-frozen GHXTox decision threshold, consistent with the freezing protocol described in Section 2.5.4; no metric-specific or test-set-specific threshold optimization was performed.

2.6.2 Ranking and Probability-Calibration Metrics

Fixed-threshold metrics describe classification at one operating point but do not fully characterize sample ranking across all possible thresholds or determine whether the predicted probabilities have reliable numerical meaning. We therefore additionally report AUROC, AUPRC, the Brier score, and ECE-10. AUROC and AUPRC evaluate probability ranking, whereas the Brier score and ECE evaluate probabilistic error and calibration.

ROC curve and AUROC. For a varying probability threshold t, the true-positive rate and the false-positive rate are

$$\mathrm{TPR}(t) = \frac{\mathrm{TP}(t)}{\mathrm{TP}(t) + \mathrm{FN}(t)}, \qquad \mathrm{FPR}(t) = \frac{\mathrm{FP}(t)}{\mathrm{FP}(t) + \mathrm{TN}(t)}. \tag{91}$$

The ROC curve plots $\mathrm{TPR}(t)$ against $\mathrm{FPR}(t)$ over all thresholds, and its area is

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}(u)\, du, \tag{92}$$

where u denotes the false-positive rate. AUROC measures the ability of the model to rank a randomly selected toxic peptide above a randomly selected non-toxic peptide. Values approaching one indicate strong ranking performance, whereas values near 0.5 generally indicate approximately random ranking. AUROC is independent of the fixed decision threshold and therefore complements the threshold-dependent metrics. Although AUROC is insensitive to class prevalence, it may underemphasize positive-class retrieval quality and the practical false-positive burden under severe class imbalance. AUPRC is therefore reported as a complementary ranking metric. Recent studies in peptide toxicity prediction commonly report ROC and PR analyses together [2–4, 7].

Precision–Recall curve and AUPRC. For threshold t, Precision and Recall are defined as

$$\mathrm{Precision}(t) = \frac{\mathrm{TP}(t)}{\mathrm{TP}(t) + \mathrm{FP}(t)}, \qquad \mathrm{Recall}(t) = \frac{\mathrm{TP}(t)}{\mathrm{TP}(t) + \mathrm{FN}(t)}. \tag{93}$$

The PR curve describes the trade-off between positive-prediction purity and toxic-peptide recall as the threshold varies. In the current implementation, AUPRC is calculated as average precision through `average_precision_score`. Let P_n and R_n denote Precision and Recall at the n-th score threshold after samples have been ranked in descending order; the reported value is

$$\mathrm{AUPRC}_{\mathrm{reported}} \coloneqq \mathrm{AP} = \sum_n (R_n - R_{n-1})\, P_n. \tag{94}$$

Throughout this study, the metric is reported as AUPRC, while its numerical value is computed using average precision. This definition weights Precision by the increase in Recall and is not necessarily identical to a trapezoidal numerical integral of the PR curve. AUPRC focuses on retrieval of the positive toxic class and is particularly informative when toxic peptides are relatively rare. Because the chance-level PR baseline is determined by the positive-class prevalence, AUPRC values for Test1 and Test2 should be interpreted within their respective class distributions rather than treated as directly comparable measures of dataset difficulty. AUROC and AUPRC are reported only when both classes are present in the evaluation subset; otherwise, the implementation returns an undefined value.

Brier score. The Brier score is the mean squared difference between the predicted probability and the binary ground-truth label [30]:

$$\mathrm{Brier} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2 . \tag{95}$$

Its range is $0 \le \mathrm{Brier} \le 1$, with lower values indicating better probabilistic predictions. The Brier score penalizes both incorrect predictions and excessive confidence: assigning a probability

of 0.01 to a truly toxic peptide incurs a larger Brier error than assigning a probability of 0.40. The Brier score reflects both probability calibration and discriminative performance and should not be interpreted as a pure calibration measure; it is therefore considered jointly with AUROC, AUPRC, and ECE.

Ten-bin expected calibration error. To further quantify the agreement between predicted confidence and observed toxicity frequency, we calculate the expected calibration error using ten equal-width probability bins, denoted ECE-10 [31]. The interval $[0, 1]$ is divided into $M = 10$ bins, and for each non-empty bin the mean predicted confidence and the observed toxic-peptide frequency are

$$\begin{aligned} \mathcal{B}_m &= \left\{ i : p_i \in \left[\frac{m-1}{M},\ \frac{m}{M} \right) \right\}, \quad m = 1, \ldots, M, \\ \mathrm{conf}\left(\mathcal{B}_m\right) &= \frac{1}{|\mathcal{B}_m|} \sum_{i \in \mathcal{B}_m} p_i, \\ \mathrm{freq}\left(\mathcal{B}_m\right) &= \frac{1}{|\mathcal{B}_m|} \sum_{i \in \mathcal{B}_m} y_i, \end{aligned} \tag{96}$$

with the final bin including $p_i = 1$. ECE-10 is defined as

$$\mathrm{ECE}_{10} = \sum_{m=1}^{10} \frac{|\mathcal{B}_m|}{N} \left| \mathrm{conf}\left(\mathcal{B}_m\right) - \mathrm{freq}\left(\mathcal{B}_m\right) \right|. \tag{97}$$

Values closer to zero indicate better agreement between predicted confidence and observed toxicity frequency; for example, a well-calibrated group of predictions near 0.8 should contain approximately 80% toxic peptides. Because ECE depends on the binning scheme, the number and boundaries of the bins remain fixed across models and datasets.

Joint interpretation of the metrics. No individual metric is sufficient to characterize peptide-toxicity prediction; the reported measures are therefore interpreted jointly. ACC summarizes overall correctness; SN and SP quantify toxic-peptide misses and non-toxic false positives, respectively; BACC evaluates class-balanced recognition; MCC provides a comprehensive correlation-based measure using all four confusion-matrix terms; AUROC and AUPRC assess ranking performance; and the Brier score and ECE-10 characterize probabilistic reliability. Exact numerical tables are used as the primary form of model comparison, with ROC and PR curves presented together. Radar charts, when included, are treated only as secondary summaries and do not replace exact metric tables or uncertainty analysis.

2.7 Robustness, Uncertainty, and Statistical Analysis

A single performance point estimate cannot fully characterize the reliability of a deep learning predictor. Model performance may be affected by random initialization, finite evaluation samples, class imbalance, predicted- structure quality, and distribution shift. GHXTox is therefore evaluated from five complementary perspectives: multi-seed reproducibility, class-stratified bootstrap confidence intervals, paired significance testing between frozen models, subgroup and structure-quality analysis, and frozen external-transfer evaluation.

All analyses in this section are conducted after the model architecture, checkpoints, probability-calibration parameters, fusion weights, and decision threshold have been frozen. No ProtT5, Mamba, EGNN, chemical-site, or conformer-residual parameters are updated during statistical evaluation, and the threshold is not modified according to the resulting confidence intervals or significance tests. This design explicitly separates model development from frozen-model evaluation.

Three sources of uncertainty are distinguished. Multi-seed analysis characterizes variation arising from stochastic optimization; bootstrap intervals quantify finite-sample uncertainty conditional on a frozen predictor; and subgroup analyses examine whether predictive performance is

concentrated within particular sequence-length, physicochemical, or structure-confidence ranges. These analyses provide complementary evidence and should not be interpreted as interchangeable measures of uncertainty.

2.7.1 Multi-Seed Reproducibility and Prediction Dispersion

Neural-network training may vary because of parameter initialization, mini-batch ordering, dropout masks, and mixed-precision computation. To assess the sensitivity of the 1D expert to these factors, three LoRA-ProtT5 and bidirectional Mamba models are trained using the random seeds $\mathcal{R} = \{42, 123, 2025\}$, while keeping the data partitions, architecture, training duration, and all hyperparameters unchanged. The current 1D expert uses an equal-weight probability average of the three full-data models rather than selecting the best-performing seed post hoc. The top-level model retains a single frozen 3D structural expert: the evaluated multi-checkpoint 3D ensemble did not satisfy the prespecified promotion criterion and was therefore not adopted in the current default GHXTox model.

For an evaluation metric M, let $M^{(s)}$ denote the result obtained using seed s. The multi-seed mean and the sample standard deviation are

$$\bar{M} = \frac{1}{S}\sum_{s=1}^{S} M^{(s)}, \quad \mathrm{SD}(M) = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}\left(M^{(s)} - \bar{M}\right)^2}, \qquad S = 3, \tag{98}$$

and seed-level reproducibility may therefore be reported as $\bar{M} \pm \mathrm{SD}(M)$. All seed-specific models must be evaluated using the same frozen threshold, or using thresholds selected without access to the corresponding test set; a test-specific optimal threshold must not be selected separately for each seed.

Prediction dispersion can also be quantified at the sample level. For peptide i, let $p_{i,1\mathrm{D}}^{(s)}$ denote the uncalibrated probability from seed s. The seed-averaged probability and the seed dispersion are

$$\bar{p}_{i,1\mathrm{D}}^{\mathrm{raw}} = \frac{1}{S}\sum_{s=1}^{S} p_{i,1\mathrm{D}}^{(s)}, \quad u_{i,\mathrm{seed}} = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}\left(p_{i,1\mathrm{D}}^{(s)} - \bar{p}_{i,1\mathrm{D}}^{\mathrm{raw}}\right)^2}, \tag{99}$$

where $\bar{p}_{i,1\mathrm{D}}^{\mathrm{raw}}$ denotes the uncalibrated three-seed average used at inference, which is distinguished from the training-time OOF average $\bar{p}_{i,1\mathrm{D}}^{\mathrm{OOF}}$ defined in Section 2.5.1. A large $u_{i,\mathrm{seed}}$ indicates that the prediction is sensitive to optimization randomness and may be used to flag boundary cases or expert-disagreement samples. It is, however, an empirical measure of training variability rather than a Bayesian posterior uncertainty estimate or a direct probability of prediction error.

The final 1D pathway follows the order defined in Section 2.3.4:

$$\left\{p_{i,1\mathrm{D}}^{(s)}\right\}_{s=1}^{3} \longrightarrow \bar{p}_{i,1\mathrm{D}}^{\mathrm{raw}} \longrightarrow \text{OOF-fitted Platt calibration} \longrightarrow p_{i,1\mathrm{D}}^{\mathrm{cal}}.$$

Seed-specific metrics characterize training stability, whereas the calibrated three-seed average is used for top-level 1D/3D fusion. The best individual seed is not substituted for the expected multi-seed result. The final calibrated ensemble prediction is a deterministic function of the three trained seed models; seed-level variability is summarized by $u_{i,\mathrm{seed}}$ rather than being treated as zero. Accordingly, variability is reported across the constituent seed models rather than by assigning a standard deviation of zero to their averaged prediction.

2.7.2 Class-Stratified Bootstrap Confidence Intervals

Performance metrics calculated on a finite test set are themselves estimates. This issue is particularly important for Test2, where a small number of changes in TP or FN can substantially

alter SN, BACC, and MCC. We therefore use class-stratified percentile bootstrap resampling to calculate 95% confidence intervals [32].

Let the positive- and negative-class index sets be

$$\begin{aligned}
&\mathcal{I}_+ = \{i : y_i = 1\}, && \mathcal{I}_- = \{i : y_i = 0\},\\
&\mathbf{i}_+^{(b)} \sim \text{SampleWithReplacement}\left(\mathcal{I}_+,\ |\mathcal{I}_+|\right), && \mathbf{i}_-^{(b)} \sim \text{SampleWithReplacement}\left(\mathcal{I}_-,\ |\mathcal{I}_-|\right),\\
&\mathbf{i}^{(b)} = \text{Concat}\left(\mathbf{i}_+^{(b)},\ \mathbf{i}_-^{(b)}\right),
\end{aligned} \tag{100}$$

where the bootstrap index vectors preserve repeated indices. This procedure preserves the original number of positive and negative samples in every replicate, avoiding extreme class-ratio fluctuations or single-class replicates when the positive class is small. The validation implementation independently resamples the positive and negative index sets with replacement.

For metric M, the bootstrap estimate is

$$M^{(b)} = M\left(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{\mathbf{i}^{(b)}}\right), \qquad b = 1, \ldots, B. \tag{101}$$

The primary test-set analysis used $B = 5000$ bootstrap replicates, whereas subgroup-difference analyses used 1000 replicates. The percentile confidence interval is

$$\text{CI}_{95\%}(M) = \left[Q_{0.025}\left(\left\{M^{(b)}\right\}_{b=1}^{B}\right),\ Q_{0.975}\left(\left\{M^{(b)}\right\}_{b=1}^{B}\right)\right], \tag{102}$$

where Q_α is the empirical α-quantile.

For ACC, BACC, SN, SP, F1, and MCC, each bootstrap replicate uses the original frozen model threshold; the threshold is not reoptimized within any resampled data set. AUROC, AUPRC, and the Brier score are calculated directly from the resampled continuous probabilities.

These intervals quantify finite-sample uncertainty conditional on the frozen predictor and the observed evaluation distribution; they do not incorporate training-seed variability or distribution shift. Bootstrap uncertainty is therefore interpreted separately from the multi-seed analysis in Section 2.7.1 and the external-transfer analysis in Section 2.7.5.

2.7.3 Paired Bootstrap and Randomization Significance Testing

When the current GHXTox model is compared with the previous default or another frozen internal baseline, both models produce predictions for the same peptides; the sample-level predictions are therefore paired. Comparing two independent confidence intervals would ignore this dependence. We instead analyze the metric difference directly on aligned samples.

Let M_A and M_B denote the metric values of frozen models A and B, respectively, where B is the current model and A is the frozen baseline. The observed difference is

$$\Delta_M^{\text{obs}} = M_B - M_A. \tag{103}$$

For BACC, SN, SP, F1, MCC, AUROC, and AUPRC, a positive difference favors model B; for lower-is-better measures such as the Brier score and ECE, a negative difference indicates improvement.

Paired class-stratified bootstrap. For every bootstrap replicate, the same class-stratified index vector $\mathbf{i}^{(b)}$ defined in Section 2.7.2 is applied to both models:

$$\Delta_M^{(b)} = M\left(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{B,\mathbf{i}^{(b)}}\right) - M\left(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{A,\mathbf{i}^{(b)}}\right), \tag{104}$$

where using identical indices preserves the paired sample structure. The 95% percentile interval is

$$\text{CI}_{95}\left(\Delta_M\right) = \left[Q_{0.025}\left(\left\{\Delta_M^{(b)}\right\}_{b=1}^{B}\right),\ Q_{0.975}\left(\left\{\Delta_M^{(b)}\right\}_{b=1}^{B}\right)\right]. \tag{105}$$

For a higher-is-better metric, an interval entirely above zero supports a consistent positive difference; for a lower-is-better metric such as the Brier score, improvement corresponds to an interval entirely below zero. Using a metric-direction coefficient s_M, where $s_M = +1$ for higher-is-better metrics and $s_M = -1$ for lower-is-better metrics, the bootstrap probability of improvement is

$$P_{\text{boot}}^{\text{improve}} = \frac{1}{B} \sum_{b=1}^{B} \mathbb{I}\left(s_M \, \Delta_M^{(b)} > 0\right). \tag{106}$$

This quantity is the proportion of bootstrap replicates in which the current model outperforms the baseline; it is not a classical hypothesis-test p-value and should not be reported using an undifferentiated P label.

Paired randomization test. A within-sample exchange randomization test is additionally used to evaluate the null hypothesis that the two frozen model outputs are exchangeable for each peptide [33]. For randomization replicate r, $z_i^{(r)} \sim \text{Bernoulli}(0.5)$ is independently drawn for each sample; if $z_i^{(r)} = 1$, the probabilities and frozen-threshold decisions of models A and B are exchanged for sample i, and otherwise they remain unchanged. The resulting null difference is

$$\Delta_{M,0}^{(r)} = M_B^{(r)} - M_A^{(r)}. \tag{107}$$

A two-sided paired randomization test was performed by comparing the absolute randomized metric differences with the absolute observed difference:

$$p_{\text{rand}} = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\left(\left|\Delta_{M,0}^{(r)}\right| \geq \left|\Delta_M^{\text{obs}}\right|\right)}{R+1}. \tag{108}$$

The primary paired analysis uses $B = 5000$ and $R = 5000$ replicates with a fixed seed of 2026. The implementation exchanges both continuous scores and the corresponding frozen-threshold decisions within each sample.

MCC and BACC are treated as prespecified primary summary measures, whereas SN, SP, AUROC, AUPRC, and the Brier score provide complementary evidence. Because several correlated metrics are examined for the same model comparison, secondary p-values are interpreted descriptively rather than as multiple independent discoveries.

Finally, Test1 and Test2 have been repeatedly observed during the historical development of the project. Paired confidence intervals and randomization p-values on these sets therefore provide retrospective evidence for differences between frozen models, rather than fully independent prospective confirmation; this limitation is explicitly recorded by the project validation implementation.

2.7.4 Subgroup and Structure-Quality Analysis

Aggregate test metrics may conceal performance heterogeneity across peptide lengths, physicochemical properties, and structure-quality ranges. This issue is especially important for structure-dependent models: an overall gain may be concentrated among high-pLDDT peptides and may not generalize to short peptides, uncertain structures, or samples with incomplete chemical-site coverage. We therefore conduct prespecified subgroup analyses without modifying the frozen model or threshold.

The primary current-model length analysis uses 11–15, 16–30, and 31–50 residues. General structural diagnostics additionally stratify samples by mean pLDDT (<0.55, 0.55–0.70, and ≥ 0.70), net charge (negative, neutral, and positive), mean hydropathy (<0, 0–1, and ≥ 1), cysteine count (0, 1–2, and ≥ 3), and chemical-site coverage (<0.80, 0.80–0.95, and ≥ 0.95). Chemical-site coverage is defined as the fraction of residues containing at least one valid chemical site:

$$r_{\text{site}} = \frac{1}{L} \sum_{i=1}^{L} \mathbb{I}\left(\text{residue } i \text{ contains at least one valid chemical site}\right). \tag{109}$$

The validation implementation also records the minimum pLDDT, the low-pLDDT residue fraction, the net charge, the mean hydropathy, and the cysteine count.

To reduce unstable conclusions from extremely small subsets, a stratum is reported only when $N_{\text{stratum}} \geq 20$ and $N_{+,\text{stratum}} \geq 5$ and $N_{-,\text{stratum}} \geq 5$. Strata that failed these reporting criteria were retained in the audit output with their sample counts and the reason for omission, but were excluded from metric-difference interpretation. These requirements match the current subgroup-validation implementation.

For each eligible stratum, BACC, SN, SP, F1, and MCC are calculated using the same threshold frozen on the full training OOF predictions; AUROC, AUPRC, and the Brier score are computed from the original continuous probabilities. No stratum-specific threshold optimization is performed.

When comparing the current model with a frozen baseline, the subgroup difference is

$$\Delta M_g = M_{B,g} - M_{A,g}, \tag{110}$$

where g denotes a stratum. Class-stratified bootstrap intervals are calculated for the MCC and AUPRC differences, using 1000 subgroup resamples in the current validation protocol.

Subgroup results are treated as heterogeneity diagnostics rather than new model-selection evidence. Because multiple variables and bins are examined, isolated positive or negative findings are interpreted together with the sample size, the positive-class count, the neighboring-stratum trends, and the confidence intervals.

In particular, pLDDT is a confidence estimate produced by a structure-prediction model. Better performance in a high-pLDDT subgroup does not establish correspondence to experimentally correct structures, and poorer performance at low pLDDT does not imply that the corresponding residues lack biological relevance. GHXTox uses continuous confidence weighting rather than excluding low-confidence samples; subgroup analysis is used to characterize applicability boundaries rather than to justify hard filtering.

2.8 Interpretability Analysis Protocol

Interpretability analyses are defined on the frozen model and do not participate in feature selection, threshold selection, or fusion-weight optimization. Unless otherwise stated, probability changes are computed with all non-target inputs held fixed, summarized separately for toxic and non-toxic peptides, and accompanied by peptide-level bootstrap intervals. The analyses therefore describe model sensitivity and expert complementarity; they are not interpreted as evidence of a causal toxicophore.

2.8.1 Residue Masking and Integrated-Gradient Attribution

For the sequence expert, each residue is first replaced in turn by the tokenization pipeline's neutral masking representation, and the change in the toxic-class probability is recorded. We complement this perturbation analysis with Integrated Gradients (IG) from a padding-compatible zero-embedding baseline to the observed ProtT5 representation [34]. Attributions are summed over embedding channels, normalized within each peptide, and then averaged across the three sequence-model seeds. We report both the signed mean and the across-seed agreement so that a large attribution from one unstable checkpoint is not presented as a robust residue signal.

2.8.2 Chemical-Site Deletion Analysis

For the structural expert, each modeled chemical site, site category, or geometric interaction is deleted individually while the remaining atom graph, coordinates, and confidence values are unchanged. The resulting probability difference quantifies the model's reliance on that candidate

interaction. Deletion scores are aggregated by functional-group class, sequence position, spatial scale, and pLDDT stratum. Because these edges encode geometrically plausible candidate relations rather than experimentally confirmed bonds, the analysis is described as model-based chemical-site sensitivity.

2.8.3 Expert Agreement and Conflict Analysis

Each benchmark peptide is assigned to one of four predeclared groups: both experts correct, only the 1D expert correct, only the 3D expert correct, or both experts incorrect. We compare the sequence probability, structural probability, fused probability, peptide length, class label, and structural confidence across these groups. Cases rescued by the 3D expert are inspected alongside cases harmed by it, preventing selective presentation of favorable examples. This analysis is performed after freezing all model parameters and is used to explain the observed complementarity rather than to retune the fusion rule on Test1 or Test2.

3 RESULTS AND DISCUSSION

3.1 Overall Predictive Performance on the Benchmark Test Sets

Using the OOF-frozen weight and threshold without dataset-specific retuning, GHXTox correctly classified 1,067 of 1,126 Test1 peptides (TP/TN/FP/FN = 287/780/26/33). This corresponds to ACC 0.9476, BACC 0.9323, SN 0.8969, SP 0.9677, F1 0.9068, MCC 0.8705, AUROC 0.9811, and AUPRC 0.9674. On the strongly imbalanced Test2 set, GHXTox correctly classified 551 of 582 peptides (35/516/20/11), yielding ACC 0.9467, BACC 0.8618, SN 0.7609, SP 0.9627, F1 0.6931, MCC 0.6673, AUROC 0.9433, and AUPRC 0.8046. The lower Test2 BACC and F1, despite high ACC, reflect the difficulty of recovering the 46 toxic peptides and demonstrate why class-balanced and precision–recall metrics are essential for this benchmark.

3.1.1 Comparison with Existing Peptide-Toxicity Predictors

Table 3 compares models reported on the same benchmark partitions. Values for ToxMSRC and ToxPLTC are transcribed from the corresponding publications or their accompanying result tables [2,3]; NR denotes an unreported metric. GHXTox obtains the highest Test1 ACC, BACC, SP, MCC, AUROC, and AUPRC among the compared rows, while ToxPLTC has higher Test1 SN. On Test2, GHXTox obtains the highest ACC, SP, MCC, and reported AUPRC, but its BACC and SN remain below both reference models. The supported conclusion is therefore a stronger overall balance of specificity, correlation, and positive-class ranking, not comprehensive superiority on every metric. Because Test1 and Test2 had been observed historically, these differences remain retrospective benchmark evidence and require confirmation on a new prefrozen external set.

Table 3: Performance comparison on the benchmark Test1 and Test2 sets.

Dataset	Model	ACC	BACC	SN	SP	MCC	AUROC	AUPRC
Test1	ToxMSRC	0.9405	0.9217	0.8781	0.9653	0.8520	0.9650	NR
	ToxPLTC	0.9378	0.9302	**0.9125**	0.9479	0.8496	0.9709	0.9426
	GHXTox (current)	**0.9476**	**0.9323**	0.8969	**0.9677**	**0.8705**	**0.9811**	**0.9674**
Test2	ToxMSRC	0.9416	0.8689	0.7826	0.9552	0.6550	0.9430	NR
	ToxPLTC	0.9261	**0.8804**	**0.8261**	0.9347	0.6197	0.9295	0.7353
	GHXTox (current)	**0.9467**	0.8618	0.7609	**0.9627**	**0.6673**	**0.9433**	**0.8046**

3.1.2 Discrimination, Calibration, and Decision Characteristics

The final fused probabilities have Brier scores of 0.0410 and 0.0436 and ECE-10 values of 0.0196 and 0.0495 on Test1 and Test2, respectively. These values indicate good overall probability accuracy, although the larger Test2 ECE suggests less stable calibration under severe class imbalance. Test2 AUPRC (0.8046) is therefore particularly informative: unlike AUROC, it directly reflects the precision–recall trade-off for the rare toxic class. The 11 false negatives on Test2 also delimit the current operating point; the model's high specificity should not be interpreted as complete minority-class coverage.

Figure 3: Overall discrimination and calibration performance of GHXTox. (A, B) Test1 ROC and PR curves; (C, D) Test2 ROC and PR curves; (E, F) Test1 and Test2 reliability diagrams.

3.2 Complementarity of the 1D and 3D Experts

The two experts are fused only after independent probability generation. The 1D branch provides the stronger standalone ranking signal, whereas the 3D branch contributes confidence-weighted geometry and local chemical context. This separation makes the gain attributable to complementary probability evidence rather than duplicated use of an earlier fused prediction.

3.2.1 Performance of the Individual Experts

Table 4 reports threshold-free discrimination and probability error for each expert. Platt scaling preserves the 1D ranking but reduces its Brier score. Relative to calibrated 1D alone, final fusion improves Test1 AUROC/AUPRC from 0.9799/0.9644 to 0.9811/0.9674 and reduces Brier from 0.0429 to 0.0410. On Test2, the corresponding values improve from 0.9409/0.7831/0.0451 to 0.9433/0.8046/0.0436. Thus, the 3D expert is weaker alone but adds useful residual ranking information when conservatively weighted.

Table 4: Performance of the individual experts and the final probability-level fusion.

Dataset	Probability stream	AUROC	AUPRC	Brier
Test1	1D, uncalibrated	0.9799	0.9644	0.0445
	1D, OOF Platt calibrated	0.9799	0.9644	0.0429
	SE3D structure expert	0.9711	0.9474	0.0587
	Equal fusion (0.50/0.50)	0.9804	0.9658	0.0429
	OOF-frozen fusion (0.72/0.28)	**0.9811**	**0.9674**	**0.0410**
Test2	1D, uncalibrated	0.9409	0.7831	0.0491
	1D, OOF Platt calibrated	0.9409	0.7831	0.0451
	SE3D structure expert	0.9295	0.7241	0.0639
	Equal fusion (0.50/0.50)	0.9428	0.7985	0.0461
	OOF-frozen fusion (0.72/0.28)	**0.9433**	**0.8046**	**0.0436**

3.2.2 Gain from OOF-Frozen Probability Fusion

The 0.72 sequence weight is an empirical result of the training-OOF search, not a manually chosen claim that sequence information is intrinsically more important. Relative to equal weighting, the frozen fusion improves AUROC/AUPRC and Brier by $0.0007/0.0016/-0.0020$ on Test1 and $0.0005/0.0061/-0.0025$ on Test2. The same full-precision threshold of 0.47022256531284556 is then applied unchanged to both benchmarks. These small but concordant changes support conservative structural weighting while also showing that most discriminative information remains in the 1D expert.

3.2.3 Expert Agreement, Disagreement, and Error Correction

Expert disagreement should be interpreted at the probability level rather than as a causal statement about individual residues. The final weighting allows structural evidence to alter borderline rankings while preventing the weaker standalone 3D branch from dominating the decision. The modest effect size and the wide paired confidence intervals reported below indicate that these corrections are sample-specific and are not guaranteed to generalize to every benchmark shift.

Figure 4: Complementarity and decision changes induced by expert fusion. (A) scatter of 1D vs SE3D probabilities; (B) error-overlap diagram; (C) probability shifts; (D) corrected/regressed/unchanged counts; (E, F) representative corrected cases.

3.3 Analysis of the Task-Adaptive 1D Sequence Expert

To investigate the role of the task-adaptive 1D expert, we first compared the current sequence architecture with the earlier sequence pipeline based on pooled ProtT5 representations and a TextCNN classifier [3], and further examined task adaptation, bidirectional state-space modeling, multi-seed ensembling, and OOF-based probability calibration. Unlike approaches that use a pretrained protein language model merely as a fixed feature extractor, GHXTox preserves residue-level ProtT5 representations, adapts them to peptide toxicity prediction through LoRA, and subsequently models residue dependencies in both the N-to-C and C-to-N directions using bidirectional Mamba.

Importantly, the gain of the new 1D expert did not manifest as a uniform increase across all threshold-dependent metrics. At their respective OOF-frozen operating points, the previous ProtT5-TextCNN sequence candidate achieved BACC, MCC, AUROC, and AUPRC values of 0.9317, 0.8586, 0.9724, and 0.9436, respectively, whereas the LoRA-ProtT5–Bi-Mamba expert achieved 0.9300, 0.8562, 0.9761, and 0.9533. Thus, BACC and MCC remained broadly comparable, while AUROC and AUPRC increased by approximately 0.0037 and 0.0097, respectively.

This pattern indicates that the main advantage of the new sequence expert is not simply a more aggressive decision boundary at one fixed threshold. Instead, the task-adaptive representation improves the ordering of continuous toxicity scores, particularly the precision–recall ranking of the positive class. Such behavior provides a stronger probability stream for subsequent fusion with structural evidence and highlights why sequence-model comparisons should not rely on ACC or BACC alone.

3.3.1 Task Adaptation of ProtT5 and the Retained 1D Representation

Existing peptide-toxicity predictors commonly employ protein language models to generate fixed sequence representations. For example, StrucToxNet uses ProtT5-derived sequence features within its graph representation, whereas PeptiTox integrates ESM2 embeddings with geometry derived from predicted structures [6, 7]. A major distinction of GHXTox is that ProtT5 is not used solely as a static upstream encoder. Instead, its linear transformations are adapted to the toxicity task through LoRA, while residue-level representations are preserved for downstream sequence modeling.

This design serves two complementary purposes. First, low-rank updates allow the pretrained semantic space to adapt to peptide toxicity prediction without requiring full fine-tuning of the ProtT5 backbone. Second, peptide-level pooling is deliberately postponed: contextualized residue states are passed to the Bi-Mamba encoder, preserving position-dependent information and allowing additional modeling of local and long-range residue dependencies.

Accordingly, the observed performance difference should not be attributed to LoRA alone. We therefore interpret LoRA-ProtT5 and Bi-Mamba as a joint task-adaptive sequence encoder. The current evidence supports the more precise conclusion that task-adapted residue-level protein language model representations followed by state-space sequence modeling improve probability ranking while preserving threshold-dependent classification performance comparable to that of the previous strong sequence baseline.

3.3.2 Bidirectional State-Space Modeling of Peptide Sequences

After task-adaptive ProtT5 encoding, GHXTox employs bidirectional Mamba rather than TextCNN or immediate peptide-level pooling for secondary sequence modeling. This choice is particularly relevant to peptides because toxicity-related evidence may not be confined to a single local motif; instead, combinations of charged, hydrophobic, aromatic, or other residues may interact over distant sequence positions. Fixed-size convolutions can capture such dependencies only through progressively enlarged receptive fields, whereas a state-space model propagates contextual information through recurrent state updates.

Although peptide sequences possess a natural N-to-C direction, toxicity classification is not an autoregressive prediction task. GHXTox therefore performs both forward and backward selective scans and integrates their representations. Each residue can consequently be contextualized by information from both termini rather than forcing the classifier to adopt a single direction of information propagation.

A clear evidence boundary should nevertheless be maintained. The current artifact set does not provide a strict single-variable comparison between unidirectional and bidirectional Mamba under otherwise identical conditions. We therefore do not assign a quantitative performance gain specifically to bidirectionality. Instead, the evidence supports the complete LoRA-ProtT5–Bi-Mamba configuration as a whole through group-aware OOF evaluation and controlled candidate screening. This distinction separates an architecture-level rationale from a causal ablation claim.

3.3.3 Multi-Seed Ensembling, Calibration, and Controlled Negative Results

After the basic sequence architecture had been established, we further investigated model variance and probability reliability. The final 1D expert was not obtained by selecting the random seed with the highest test performance. Instead, three models were independently trained on the complete training set and their probabilities were averaged with equal weights. A larger ensemble constructed directly from the 15 fold-specific checkpoints did not yield additional gains. A plausible explanation is that each fold member is trained on only approximately 80% of the available training samples, and the resulting reduction in individual-model capacity outweighs the variance reduction obtained from increasing the ensemble size. The final configuration therefore uses the fixed average of three full-data models.

Following seed averaging, training OOF predictions were used to fit the monotonic Platt calibrator. This operation primarily adjusts the probability scale without altering the ranking of samples. The observed results were consistent with this role: the 1D Brier score decreased from 0.0445 to 0.0429 on Test1 and from 0.0491 to 0.0451 on Test2. After subsequent fusion with the 3D expert, the Brier scores were further reduced to 0.0410 and 0.0436, respectively. These results indicate that calibration is not merely a cosmetic post-processing step, but an important component for producing probability streams that can be meaningfully combined across experts.

Three intuitively attractive extensions were additionally evaluated but not retained. Conservative positive-class sequence perturbation shifted predictions toward the positive class and degraded specificity and the Brier score. Length-specific calibration or thresholding was unstable because of increased sampling variance within smaller strata and did not consistently outperform the global operating point. Finally, the 15-member fold ensemble did not outperform the three full-data models.

These negative results are methodologically informative. They indicate that, at the current data scale, the limitation of the sequence expert cannot be resolved simply by generating additional near-duplicate samples, increasing the ensemble size, or fragmenting the decision rule into multiple length-specific operating points. The retained three-seed full-data average with OOF calibration represents a simpler and more stable solution and reduces the risk of selective reporting or continued optimization against historically observed test sets.

Table 5: Selected controlled comparisons for the task-adaptive 1D sequence expert. OOF architecture rows and benchmark calibration rows are reported separately and should not be compared across evaluation splits.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC	Brier
Training OOF	Pooled ProtT5–TextCNN	0.9317	0.8586	0.9724	0.9436	NR
Training OOF	LoRA-ProtT5–Bi-Mamba	0.9300	0.8562	**0.9761**	**0.9533**	NR
Test1	Three-seed 1D, uncalibrated	–	–	0.9799	0.9644	0.0445
Test1	Three-seed 1D, OOF Platt	–	–	0.9799	0.9644	**0.0429**
Test2	Three-seed 1D, uncalibrated	–	–	0.9409	0.7831	0.0491
Test2	Three-seed 1D, OOF Platt	–	–	0.9409	0.7831	**0.0451**

3.4 Analysis of the pLDDT-Aware Multi-Scale 3D Structural Expert

Compared with the 1D expert, the principal methodological innovations of GHXTox lie in its structural expert. Previous studies have already demonstrated the value of predicted structures for toxicity prediction. PeptiTox combines ESM2 semantics with geometric descriptors extracted from predicted structures, whereas StrucToxNet further employs an equivariant graph neural network to preserve consistency under rigid transformations of the three-dimensional coordinates [6, 7]. For short peptides, however, a more fundamental question is not simply whether three-dimensional coordinates are available, but which structural evidence should be trusted and at which spatial scale that evidence should be used.

The GHXTox structural expert is organized around this question. ESMFold predictions are not treated as noise-free experimental structures. Instead, pLDDT acts as a continuous reliability variable throughout geometric message passing, residue-level gating, peptide-level gating, and training-time coordinate perturbation. The model then extends beyond Cα-level residue geometry by extracting local chemical pseudo-sites from full-atom PDB structures and, when conformer ensembles are available, incorporating statistics such as RMSF, distance variation, and contact occupancy through a conservative residual pathway.

The central 3D contribution is therefore not merely the addition of more structural features, but a hierarchical structural representation that explicitly organizes evidence by reliability and spatial scale–from residue geometry and local side-chain chemistry to sampled conformational variation.

3.4.1 Continuous Structural-Confidence Modeling Is Preferable to Hard Filtering

Structure-quality analysis first confirmed that pLDDT contains useful information: peptides with higher mean pLDDT generally exhibit stronger predictive performance. However, this does not imply that discarding low-confidence samples is an optimal modeling strategy. In a controlled high-pLDDT-only experiment, restricting training to high-confidence structures substantially reduced performance over the complete test domain, with MCC values of only 0.7232 on Test1 and 0.5593 on Test2.

This observation highlights an important distinction: structural confidence is more useful for modulating structural evidence than for deciding whether a sample should be retained at all. Low pLDDT may arise from genuine peptide flexibility, limited information in short sequences, or uncertainty specific to the structure-prediction model. Hard exclusion produces a structurally cleaner subset but simultaneously removes many difficult examples that remain part of the target prediction domain.

GHXTox therefore uses a continuous confidence-aware strategy. Low-confidence edges remain in the graph but contribute weaker messages; residue-level gates modulate structural evidence at individual positions; peptide-level gating controls the overall use of structural context;

and stronger coordinate perturbations are applied to low-pLDDT residues during training.

This is an important distinction from approaches that analyze structure quality only after model inference. In GHXTox, pLDDT is not merely a diagnostic variable–it participates directly in representation learning.

3.4.2 Full-Atom Chemical Sites Provide Consistent but Complementary Structural Gains

A Cα-level residue graph effectively describes peptide compactness, nonlocal contacts, and residue neighborhoods but compresses chemically complex side chains into single residue nodes. To determine whether local spatial chemistry provides additional discriminative information, we introduced a full-atom-PDB-derived chemical-site residual on top of the frozen Cα-EGNN. A strict paired five-fold design was adopted: in each fold, the candidate model was initialized from the same control checkpoint, while the sequence branch, Cα-EGNN, fusion module, and classifier were frozen, leaving only the newly introduced chemical-site branch trainable.

Across the five CD-HIT homology-grouped folds, the complete chemical-site model increased mean BACC from 0.9153 to 0.9171 ($\Delta = +0.0018$), F1 from 0.8790 to 0.8814 ($\Delta = +0.0023$), MCC from 0.8312 to 0.8342 ($\Delta = +0.0030$), AUROC from 0.9687 to approximately 0.9692, and AUPRC from 0.9388 to 0.9401 ($\Delta = +0.0013$). More importantly, MCC, F1, and AUPRC improved in all five folds.

The absolute magnitude of these gains is modest, but their consistency is methodologically more informative. Because the base geometry, sequence semantics, and classifier were frozen, the observed incremental performance can only arise from the newly introduced chemical-site pathway under this controlled training setup. The result therefore suggests that residue-centered geometry already captures most of the structural signal, but local side-chain chemistry contains a small yet reproducible component of nonredundant predictive information.

Component-level ablations further support this interpretation. On the fixed fold-0 screening subset, removing the second chemical site, orientation invariants, either distance scale, explicit compatibility channels, pLDDT weighting, or zero initialization decreased MCC. Retaining only the raw-distance or only the normalized-distance representation reduced MCC by approximately 0.0059 and 0.0058, respectively, whereas removing interaction-type channels or pLDDT weighting each reduced MCC by approximately 0.0057.

The methodological contribution is therefore not simply the construction of a larger atom-level graph. GHXTox compresses local side-chain environments into chemically interpretable pseudo-sites while simultaneously preserving physical distances in Å, peptide-size-normalized distances, orientation relationships, and candidate chemical compatibility.

PeptiTox demonstrated the utility of geometric information from predicted structures, whereas ToxiPep highlighted complementarity between local atomic chemistry and sequence context [4,6]. GHXTox integrates these ideas within a confidence-aware 3D structural expert. The resulting interaction channels should nevertheless be interpreted as candidate spatial-chemical relationships available to the model, rather than confirmed hydrogen bonds, salt bridges, cation–π interactions, or disulfide bonds.

3.4.3 Sequence–Structure Interaction and Hierarchical Confidence Gating

Local geometric evidence becomes meaningful only in the appropriate sequence context. The same spatial distance or hydrophobic contact may have different implications depending on residue identity and sequence position. GHXTox therefore avoids simply concatenating sequence and structural vectors after peptide-level pooling. Instead, contextualized sequence states act as queries that selectively retrieve evidence from pLDDT-gated spatial residue representations.

This idea is conceptually related to the context–chemistry alignment emphasized by ToxiPep, but in GHXTox the interaction occurs within a predicted three-dimensional structural

expert and is further regulated by both residue-level and peptide-level structural-confidence gates.

The residue-level gate modulates how strongly the structural representation of each residue contributes to the subsequent interaction, whereas the peptide-level gate controls the overall contribution of structural context. Structural information is therefore not switched off when confidence decreases; rather, its contribution is continuously attenuated at multiple levels.

Together with the negative result of hard high-pLDDT-only training described in Section 3.4.1, these observations suggest that the value of the retained design lies not merely in increasing model capacity, but in establishing a learnable compromise between structural-information retention and confidence-aware attenuation.

Because the current artifact set does not provide a complete single-variable ablation of cross-attention and each confidence gate under otherwise identical conditions, the overall structural gain is not quantitatively decomposed among these components. The supported conclusion is therefore that continuous pLDDT weighting, sequence–structure cross-attention, and hierarchical gating jointly constitute the retained reliability-aware interaction mechanism.

3.4.4 Conformer Statistics Provide Conditional Gains with Exact Fallback

A single ESMFold prediction provides only one deterministic structural anchor, whereas short peptides may exhibit substantial conformational heterogeneity. GHXTox does not treat every generated conformer as an additional labeled training example, nor does it repeatedly execute the complete EGNN for every conformation. Instead, conformer ensembles are compressed into statistics including RMSF, pairwise-distance variation, contact occupancy at 6, 8, and 10 Å, conformer RMSD, and radius of gyration, and these statistics are allowed to produce only a zero-initialized logit residual on top of the single-structure predictor.

Within the prespecified pilot250 subset, the conformer-statistics residual improved MCC by approximately 0.056 relative to the single-structure baseline, with a 95% confidence interval of [0.011, 0.108]; a paired permutation test yielded $p = 0.036$. When evaluated over the complete 6,387-sample training OOF set, however, the overall MCC gain decreased to approximately +0.0028, although the confidence interval remained positive at [0.0007, 0.0052], accompanied by a reduction in Brier score.

The contrast between the relatively strong pilot effect and the smaller full-data gain is itself informative. Conformer information is not equally available across samples, and conformer coverage is uneven across peptide-length ranges and toxicity classes. Additional conformers therefore do not translate linearly into improved classification performance. In the subsequent PepFlow coverage expansion, the additional 511 supported peptides were all non-toxic and did not further improve MCC.

For this reason, the conformer pathway is not implemented as a mandatory backbone. Instead, GHXTox uses an availability-masked residual. For peptides without valid conformer statistics, $\Delta z_{\text{conf}} = 0$, and the prediction therefore falls back exactly to the single-structure pathway.

The conformer module should therefore be interpreted as a conditional structural refinement mechanism rather than a universal explanation for performance gains across all benchmark samples.

3.4.5 Structural Expert Performance and the Nature of the 3D Contribution

After combining the retained components, the 3D-v2 structural expert achieved BACC, MCC, AUROC, and AUPRC values of 0.9208, 0.8464, 0.9711, and 0.9474 on Test1, compared with 0.9179, 0.8458, 0.9699, and 0.9426 for the historical Cα-only 3D-v1 model. Test1 therefore showed a consistent but modest improvement. Test2 exhibited a different trade-off: AUPRC increased from 0.6969 to 0.7241, whereas MCC decreased from 0.6320 to 0.6128.

This trade-off is particularly instructive. It indicates that finer-grained structural modeling improves positive-class probability ranking but does not guarantee simultaneous gains in all metrics at a frozen operating point. This is particularly relevant for Test2, which contains only 46 toxic peptides, where a small number of threshold-adjacent samples can substantially alter MCC and F1.

The conclusion from the structural expert is therefore not that a more complex 3D model universally outperforms its predecessor. Rather, confidence-aware multi-scale structural modeling introduces a small but consistent incremental signal under group-aware training OOF evaluation and improves probability ranking on selected historical benchmarks, while threshold-dependent gains remain dataset-dependent.

This result also clarifies the role of the structural expert within GHXTox. The 3D branch is most valuable as a source of complementary evidence rather than as a universal replacement for a strong sequence predictor, providing a direct rationale for the final dual-expert probability fusion.

Table 6: Ablation analysis of the 3D structural expert. The first four rows use historical benchmark predictions; the final two rows are a paired five-fold OOF chemical-site experiment.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC
Test1	Historical Cα 3D-v1	0.9179	0.8458	0.9699	0.9426
Test1	pLDDT-aware multi-scale 3D-v2	**0.9208**	**0.8464**	**0.9711**	**0.9474**
Test2	Historical Cα 3D-v1	**0.8481**	**0.6320**	**0.9316**	0.6969
Test2	pLDDT-aware multi-scale 3D-v2	0.8453	0.6128	0.9295	**0.7241**
Training OOF	Frozen backbone, no chemical-site residual	0.9153	0.8312	0.9687	0.9388
Training OOF	Frozen backbone + chemical-site residual	**0.9171**	**0.8342**	**0.9692**	**0.9401**

Figure 5: Module-level ablation results for the 1D and 3D experts (A–F).

3.5 Robustness, Statistical Uncertainty, and Applicability Analysis

Point estimates alone are insufficient for distinguishing a stable model improvement from variation caused by a finite benchmark sample. We therefore retain multi-seed averaging for the 1D expert and use paired, sample-level bootstrap resampling to compare the current default with the immediately preceding frozen model. All comparisons preserve the pairing between the two models' predictions for each peptide.

3.5.1 Multi-Seed Reproducibility

Three full-training 1D models initialized with seeds 42, 123, and 2025 are averaged before calibration. This ensemble is the deployed predictor; no seed is selected according to Test1 or Test2 performance. A 15-member ensemble of fold-specific checkpoints was also examined but not retained because its members had each seen only approximately 80% of the training set and the aggregate did not improve the frozen validation criteria.

3.5.2 Bootstrap Confidence Intervals

The paired bootstrap used 5,000 resamples with seed 2026. As expected from the small toxic subset in Test2, its intervals are substantially wider than those of Test1, especially for BACC and MCC. Consequently, a favorable point estimate on Test2 is not treated as proof of a population-level improvement.

3.5.3 Paired Comparison with the Previous Model

Table 7 reports changes from the previous frozen default to the current LoRA-ProtT5–Bi-Mamba/SE3D fusion. Positive values are favorable for BACC, MCC, and AUPRC, whereas negative values are favorable for Brier. The current model improves the point estimates of all four displayed metrics on both benchmarks, but every 95% interval crosses zero. These results justify promotion under the project's prespecified multi-metric retention rule, while not supporting a claim of statistically established superiority.

Table 7: Bootstrap confidence intervals and paired statistics on Test1 and Test2.

Dataset	Metric	Current–previous Δ	Paired 95% CI	$P(\Delta$ favorable)
Test1	BACC	+0.0072	[−0.0034, 0.0190]	0.9086
	MCC	+0.0093	[−0.0112, 0.0314]	0.8240
	AUPRC	+0.0024	[−0.0041, 0.0093]	0.7616
	Brier	−0.0022	[−0.0065, 0.0019]	0.8494
Test2	BACC	+0.0009	[−0.0307, 0.0326]	0.5200
	MCC	+0.0071	[−0.0524, 0.0694]	0.5810
	AUPRC	+0.0196	[−0.0235, 0.0678]	0.7860
	Brier	−0.0036	[−0.0103, 0.0027]	0.8682

3.5.4 Length-Stratified Performance and Conformer Availability

We first examined whether the change from the previous multidimensional default to the current LoRA-ProtT5–Bi-Mamba/SE3D model was consistent across peptide-length domains. Test1 and Test2 were divided into 11–15, 16–30, and 31–50 aa strata, and each model was evaluated at its own training-OOF-frozen operating point. The sample and positive-class counts are displayed directly in Fig. 6; no length-specific threshold was selected from either benchmark.

On Test1, the current model increased BACC/MCC from 0.8283/0.7518 to 0.8567/0.7746 for 11–15 aa peptides and from 0.9604/0.9091 to 0.9714/0.9292 for 31–50 aa peptides. In the 16–30 aa stratum, however, BACC/MCC decreased from 0.8982/0.8164 to 0.8940/0.8037, despite an AUPRC increase from 0.9367 to 0.9444. Test2 showed the same qualitative heterogeneity: BACC/MCC increased slightly in the 11–15 and 31–50 aa strata, whereas the 16–30 aa stratum decreased from 0.8039/0.4803 to 0.7484/0.4049. The latter stratum contains only nine toxic peptides, so one or two decision changes have a disproportionately large influence on BACC and MCC. AUPRC nevertheless increased from 0.4803 to 0.5882 in this subgroup. Thus, four of the six dataset-by-length strata improved at the frozen operating points, but the update did not provide a uniform gain across all length domains.

We then isolated the conformer-ensemble pathway by holding the current LoRA-ProtT5–Bi-Mamba probabilities, the 0.72/0.28 fusion weights, and the threshold of 0.4702225653 fixed, and replacing only the multi-structure-enabled 3D probability with the corresponding single-structure 3D probability. The two current variants produced identical BACC, MCC, and AUPRC in every Test1 and Test2 length stratum. Their maximum sample-level 3D probability differences were only 5.59×10^{-6} and 3.28×10^{-7}, respectively. These held-out outputs therefore reflect the exact single-structure fallback of the availability-masked conformer pathway, not a measurable Test1/Test2 gain from multi-conformer inputs. Accordingly, the length-stratified improvement relative to the previous default must be attributed to the complete model update and cannot be assigned specifically to conformer-ensemble enhancement.

When conformer ensembles are available, different predicted conformations of the same sequence are first aligned by the Kabsch procedure to remove global translation and rotation. GHXTox then extracts residue RMSF, conformer RMSD, radius of gyration, pair-distance mean

and variation, and contact occupancies at 6, 8, and 10 Å. Continuous geometric quantities are rescaled by fixed physical constants before residue-level and peptide-level encoding. The resulting ensemble representation is incorporated through an availability-masked logit residual and therefore forms a dedicated conformational-statistics scale within the hierarchical multi-scale 3D expert, alongside residue geometry and local chemical-site interactions. This design addresses information omitted by a single deterministic structure in the covered domain while preserving an exact single-structure fallback when the ensemble is unavailable.

3.5.5 External Evidence and Generalization Boundary

3.6 Interpretability and Representative Case Studies

3.6.1 Residue-Level Interpretation of the 1D Expert

3.6.2 Structural and Chemical-Site Interpretation

3.6.3 Expert Agreement and Conflict Interpretation

3.6.4 Representative Conformer Cases

3.7 Limitations and Future Directions

Although GHXTox establishes a comprehensive peptide-toxicity prediction framework through task-adaptive sequence modeling, reliability-aware three-dimensional structural representation, and OOF-frozen probability fusion, the current findings should be interpreted within several important boundaries related to data provenance, structure-prediction reliability, conformer coverage, fusion strategy, and the level of interpretability evidence.

Importantly, unsuccessful experiments were not discarded; non-promoted structural extensions, dynamic fusion schemes, and conformer-expansion candidates were systematically recorded. Consequently, the remaining limitations do not simply reflect the absence of additional architectural components; instead, they reveal several problems that cannot yet be resolved reliably under the current data and validation regime.

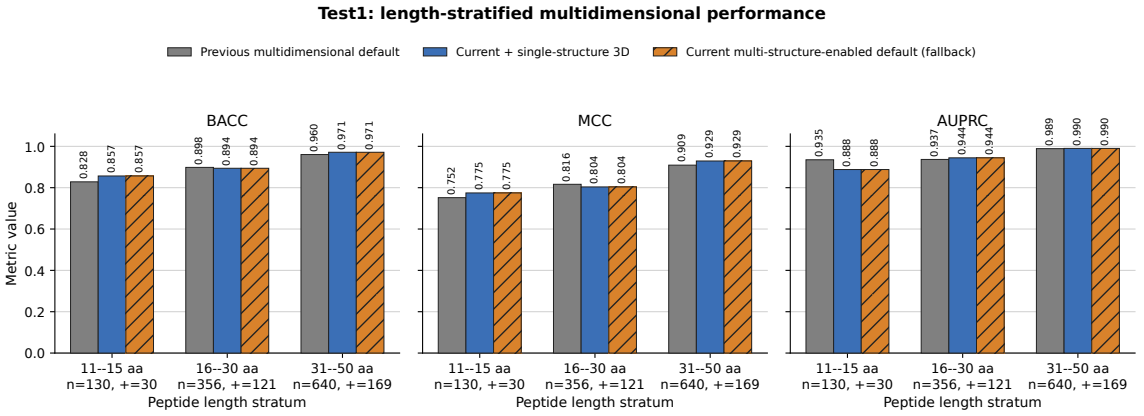
Future work should therefore not be framed merely as increasing model complexity. The more important directions include obtaining genuinely independent generalization evidence; improving conformational coverage without generator–length confounding; incorporating environment- and topology-dependent structural information; developing more rigorous uncertainty handling; and connecting model-derived attribution patterns with experimental validation.

3.7.1 Dataset and Evaluation Limitations

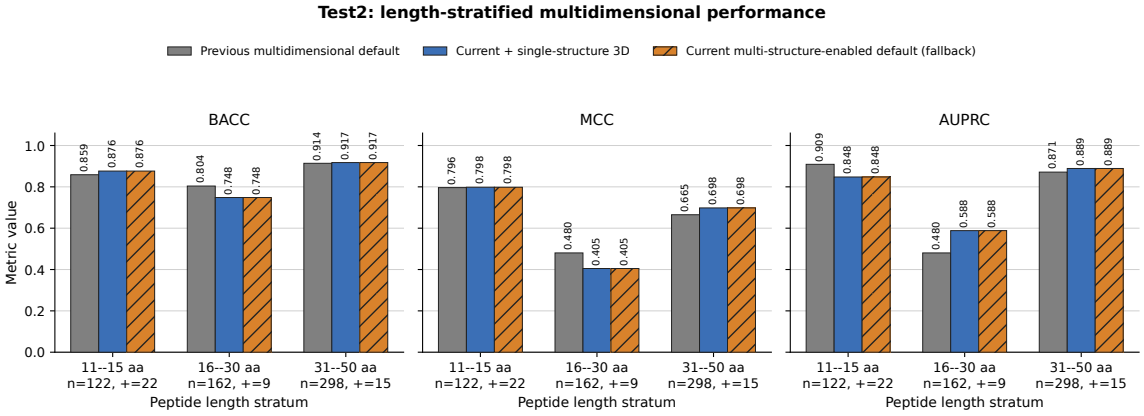
Although calibration, fusion parameters, and the decision threshold were determined using group-aware OOF predictions without any formal tuning on Test1 or Test2, both benchmark sets had been observed during historical model development. They should therefore be regarded as frozen historical benchmarks rather than newly blinded external test sets. In addition, Test2 contains only 46 toxic peptides, making SN, BACC, and MCC sensitive to a small number of prediction changes. A new external dataset evaluated under a prefrozen protocol will therefore be required to provide stronger evidence of cross-domain generalization.

3.7.2 Structural and Conformer Limitations

The 3D expert depends on ESMFold-predicted structures and is therefore subject to structure-prediction uncertainty. Although pLDDT allows structural evidence to be continuously down-weighted, it is not equivalent to experimental structural accuracy, conformational flexibility, or dynamical stability. Similarly, full-atom-derived chemical sites represent candidate



(a) Test1 length strata.



(b) Test2 length strata.

Figure 6: Length-stratified comparison of the previous multidimensional default, the current model reconstructed with the single-structure 3D expert, and the canonical multi-structure-enabled current default. BACC, MCC, and AUPRC are shown together with total and positive-class sample counts. The two current variants hold the 1D probabilities, 0.72/0.28 fusion, and current OOF-frozen threshold fixed; their overlap shows that no effective conformer residual altered the held-out Test1 or Test2 metrics.

Figure 7: Representative interpretability cases for the 1D expert, the 3D expert, and the fused prediction.

spatial-chemical relationships available to the model rather than confirmed hydrogen bonds, salt bridges, or cation–π interactions.

Conformer coverage is also uneven, with some coupling between peptide length and conformer generator. The conformer pathway should therefore be interpreted as a conditional refinement when ensemble information is available, rather than a universal source of improvement. Future work should construct same-sample, multi-generator conformer sets to better separate generator effects from peptide-length effects.

3.7.3 Modeling and Interpretability Limitations

GHXTox currently trains the two experts separately and combines them through probability-level linear fusion rather than full end-to-end joint optimization. This design provides clear training and calibration boundaries, but a fixed fusion weight cannot adapt the relative importance of sequence and structural evidence for individual peptides. Dynamic gating and stacking strategies were explored but did not consistently outperform the frozen configuration, suggesting that greater fusion flexibility may introduce additional overfitting at the current data scale.

Interpretability is also incomplete. Existing residue- and structure-level analyses mainly characterize the 3D expert, whereas matched residue-level attribution for the LoRA-ProtT5–Bi-Mamba expert has not yet been completed. Current explanations should therefore be interpreted as model-dependent evidence, rather than confirmed toxicity mechanisms or causal residues.

3.7.4 Future Directions

Future work should focus on three main directions. First, the frozen GHXTox configuration should be evaluated on a new independent external dataset to obtain stronger evidence of generalization. Second, conformer coverage should be expanded with better control of peptide length and structure generator, with future structural representations incorporating factors such as solvent environment, membrane context, protonation state, and peptide topology. Third, residue-level token occlusion or gradient attribution should be implemented for the 1D expert and compared directly with 3D structural hotspots, followed where possible by mutational or toxicity experiments.

More broadly, the next stage of GHXTox should prioritize independent validation, better-controlled structural evidence, and experimentally testable hypotheses, rather than increasingly small gains on existing benchmarks.

4 CONCLUSIONS

GHXTox combines a task-adaptive LoRA-ProtT5–Bi-Mamba sequence expert with a pLDDT-aware multi-scale structural expert through an auditable probability fusion protocol. The structural expert integrates residue geometry, full-atom-derived chemical sites, and an availability-masked conformer residual, while the final predictor uses the pure structural probability rather than recycling a previous fused output. Training-OOF predictions freeze the 1D/3D weights at 0.72/0.28 and the decision threshold at 0.4702225653128456 before benchmark evaluation.

On the frozen historical Test1 and Test2 benchmarks, the current model reaches MCC values of 0.8705 and 0.6673 and AUPRC values of 0.9674 and 0.8046, respectively. The 3D expert is weaker than the 1D expert in isolation, but its conservative inclusion improves ranking and Brier score on both sets. These results support multi-dimensional complementarity, while the paired bootstrap intervals and Test2 minority-class performance preclude claims of universal or statistically established superiority. The next decisive step is evaluation on a genuinely new,

prefrozen external dataset together with better-controlled same-sample conformer evidence and matched 1D/3D attribution analyses.

Data and Software Availability

The source code, configuration files, fold manifests, and inference scripts used in this study are available at [repository]. The manuscript results correspond to commit [HASH] and release [VERSION]. Frozen model parameters, calibration coefficients, fusion weights, the full-precision decision threshold, and sample-level prediction files are provided in [archive/release]. Raw peptide records were obtained from the public databases described in Section 2.1. Where redistribution is restricted by the original database terms, accession lists and preprocessing scripts are provided instead of redistributing the source records.

Supporting Information

Supporting Information is available and includes detailed dataset audits, fold-wise results, hyperparameter settings, additional ablation experiments, subgroup analyses, statistical outputs, and extended interpretability case studies.

CRediT Authorship Contribution Statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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